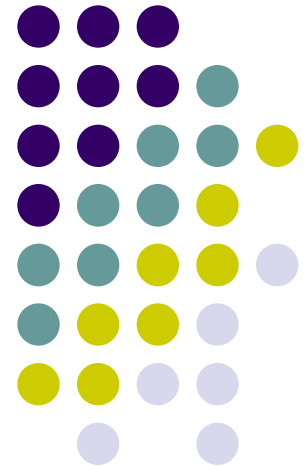
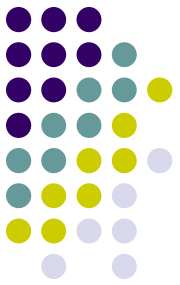


Modelling Behaviour via LeadsTo formalism

Dynamic Modelling for
Human-centered Systems

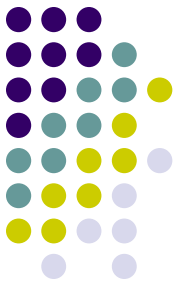
lecture 4





Course outline

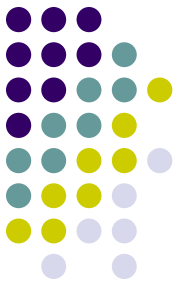
- Week 1: what is dynamic modelling?
- Previous lecture: modelling social processes
- **Today: modelling behaviour**
- Week 4: creating model from scratch
- Week 5: agent-based modelling
- Week 6: analysis and support models
- Week 7: wrap-up



Three aims for today

- Brief rehearsal of basic concepts of modelling
- How do we model behavior?
- Introduction of new logical formalism
LEADSTO (used in L2-Python for next assignments)

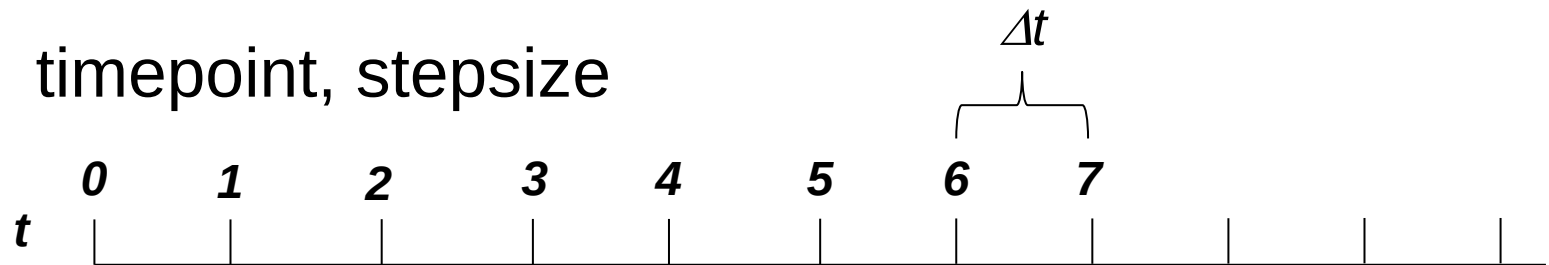
Elements of a dynamic model



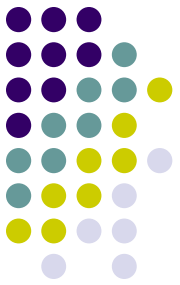
- Concepts
 - represented as variables
 - either numerical or logical (“sentences” that are true/false)

- Timeline

- timepoint, stepsize

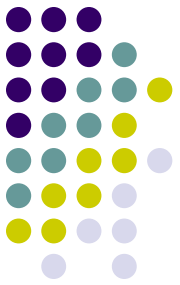


- Rules that describe how concepts change per time step



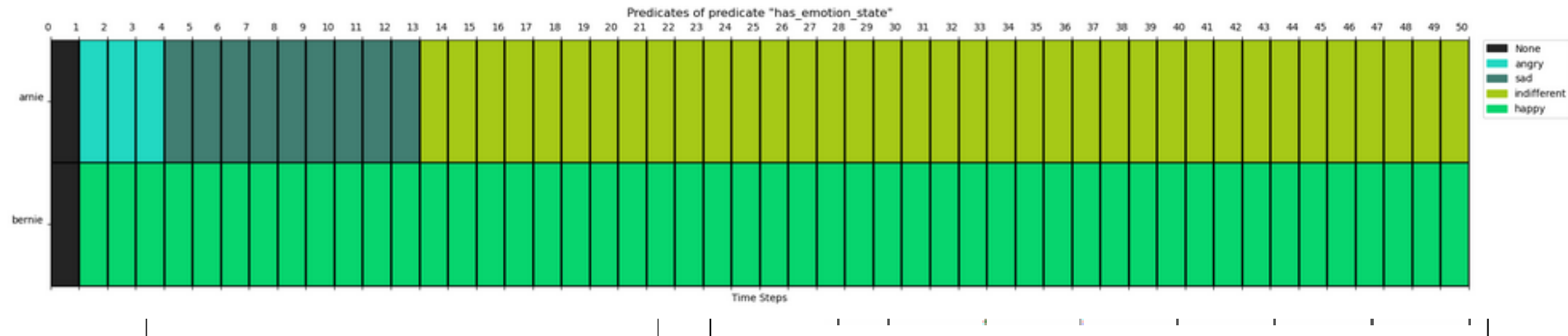
Additional elements

- Optional parameters
 - $\beta = 0,33$
- Optional stochastic variables
 - random values (not so much in logical models)
- Given values that define a specific simulation
 - a **scenario**
 - start values, but also external events that happen at some time point

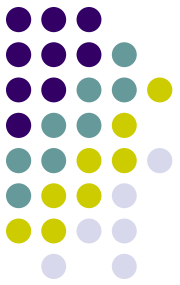


Model outcome

- List of states of a model at subsequent time points:
 - (simulation) trace
 - simulation run



Summary



scenario
 $\text{conceptA}(1) = 8$
 $\text{conceptB}(10-14) = 1$



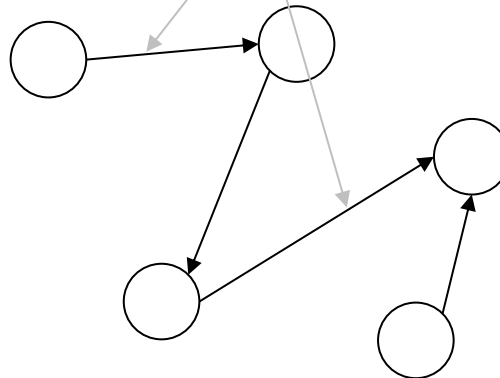
parameters

$$\beta = 0,33$$

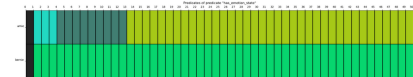
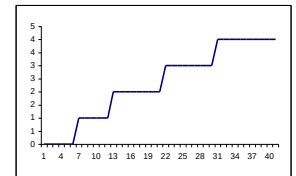
$$\alpha = 0,7$$



model



output
simulation trace

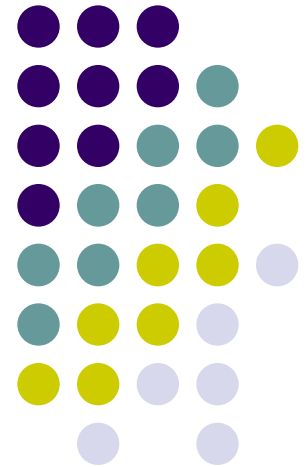


length
40 time steps

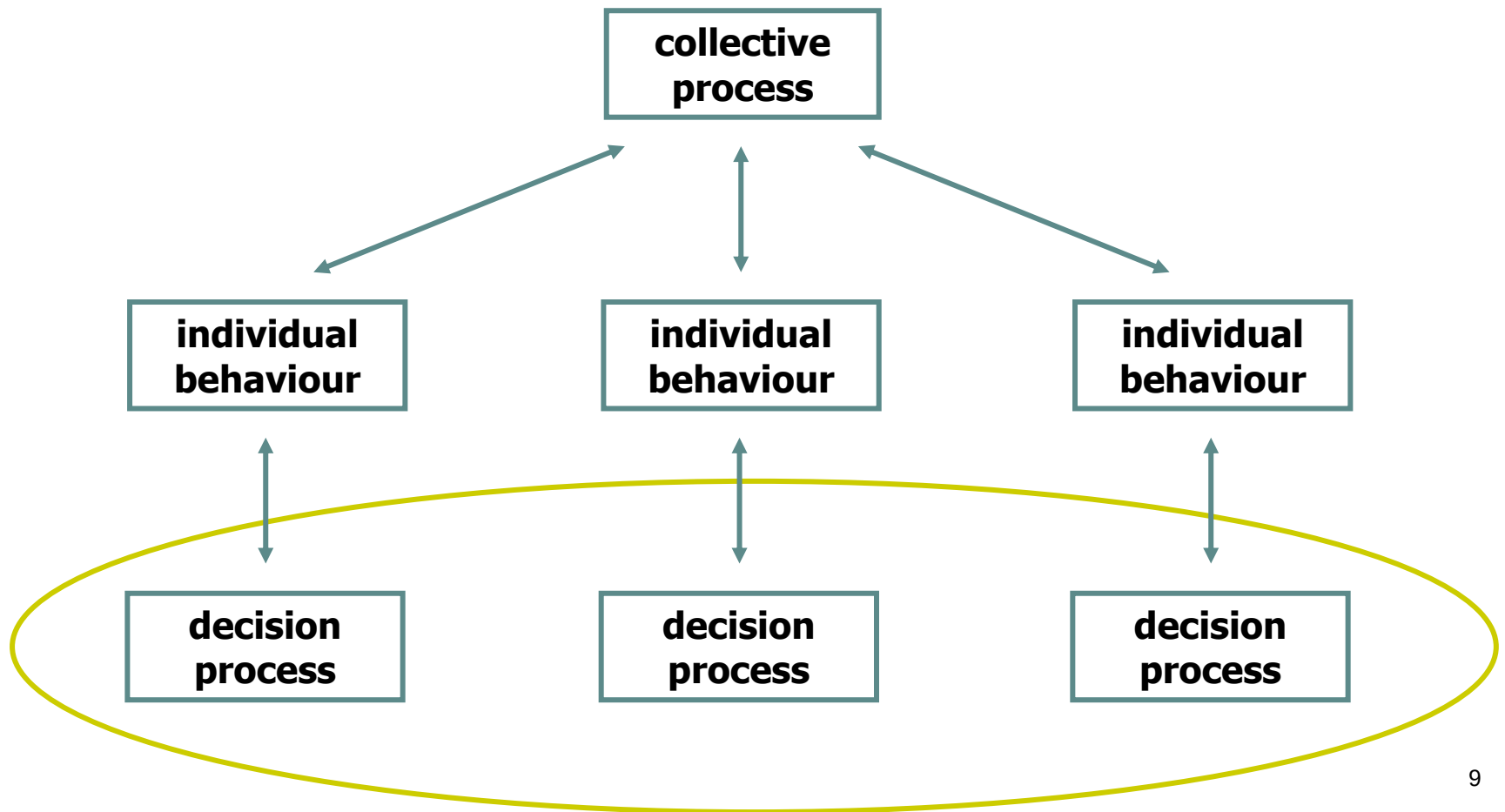
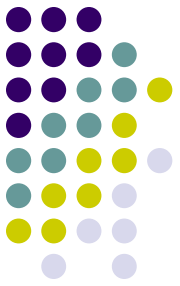
Behaviour modelling

Dynamic Modelling for
Human-centered Systems

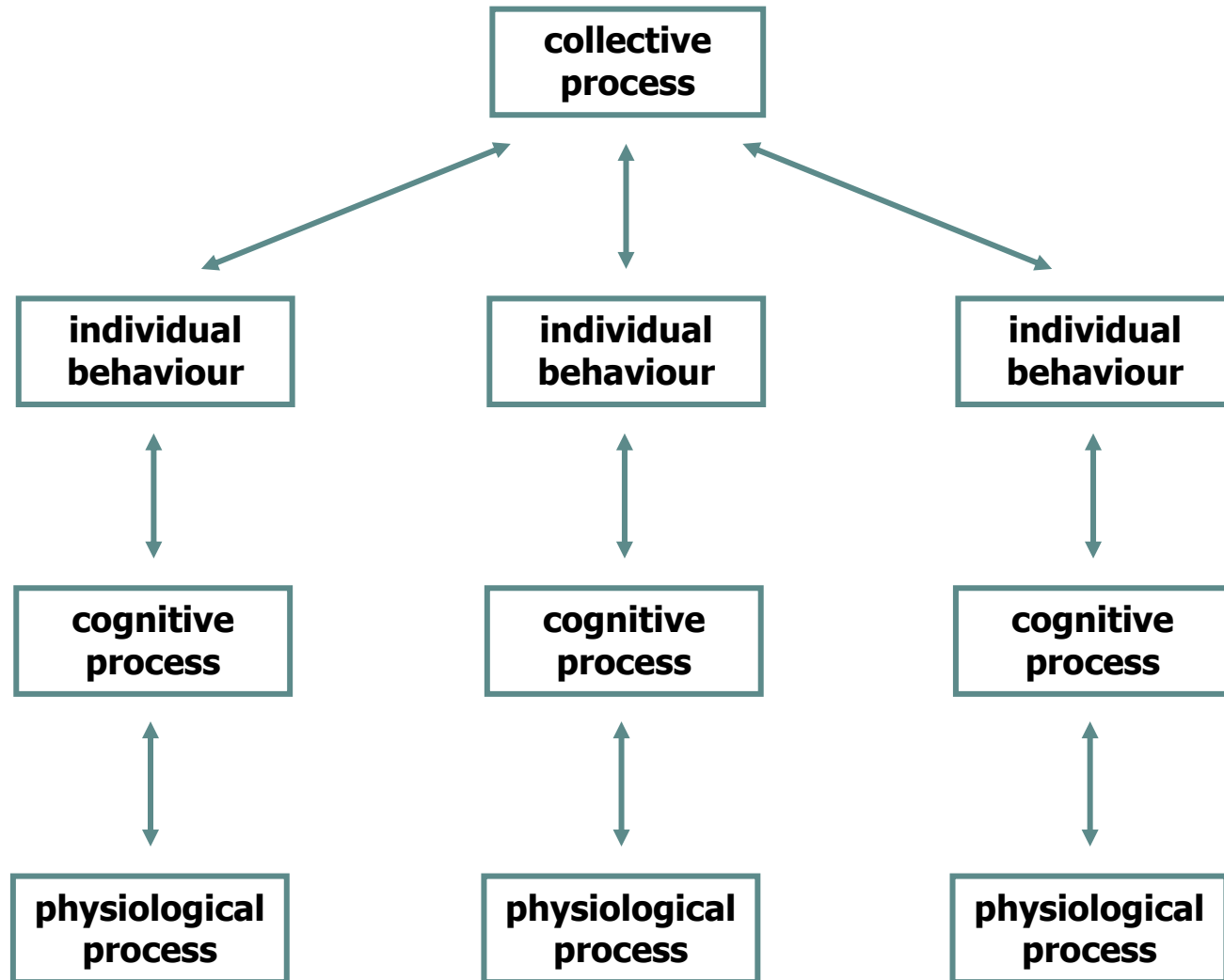
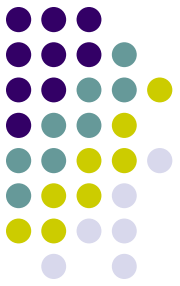
Chapter 8

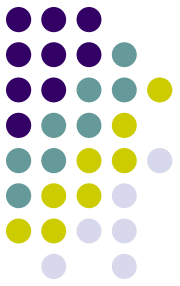


Collective pattern and individual behaviour



Levels of modelling

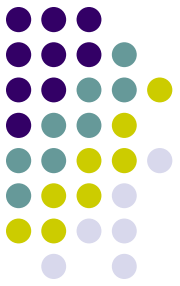




Goals of next three lectures

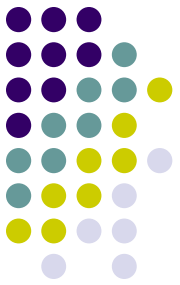
- Understanding how to model behaviour
 - relating **behavioural** level and **cognitive** level
- Learning common types of behavior
- Understanding of hybrid modelling mechanism (LEADSTO)
 - semi-formal notation
 - formal notation

Behaviour



- What is behaviour?
- How can we model (intelligent) behaviour?





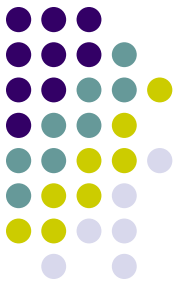
What is Behaviour?

Wikipedia:

“Behaviour is the range of **actions** and mannerisms made by organisms, systems, or artificial entities **in conjunction with their environment**, which includes the other systems or organisms around as well as the physical environment.

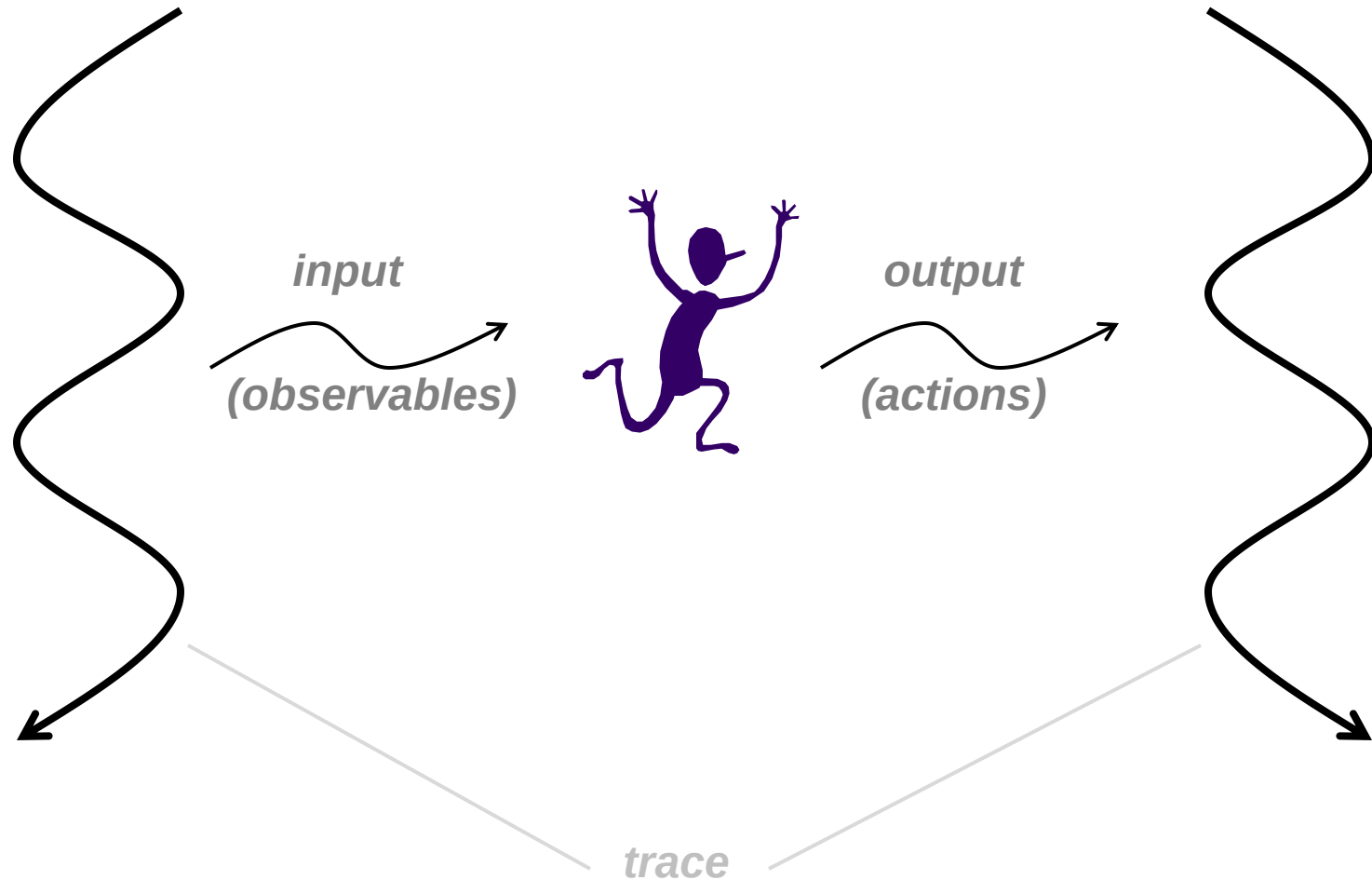
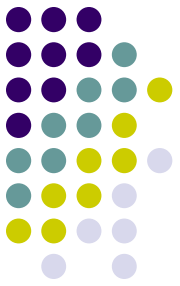
It is the response of the system or organism to various stimuli or inputs, whether internal or external, conscious or subconscious, overt or covert, and voluntary or involuntary.”

Question

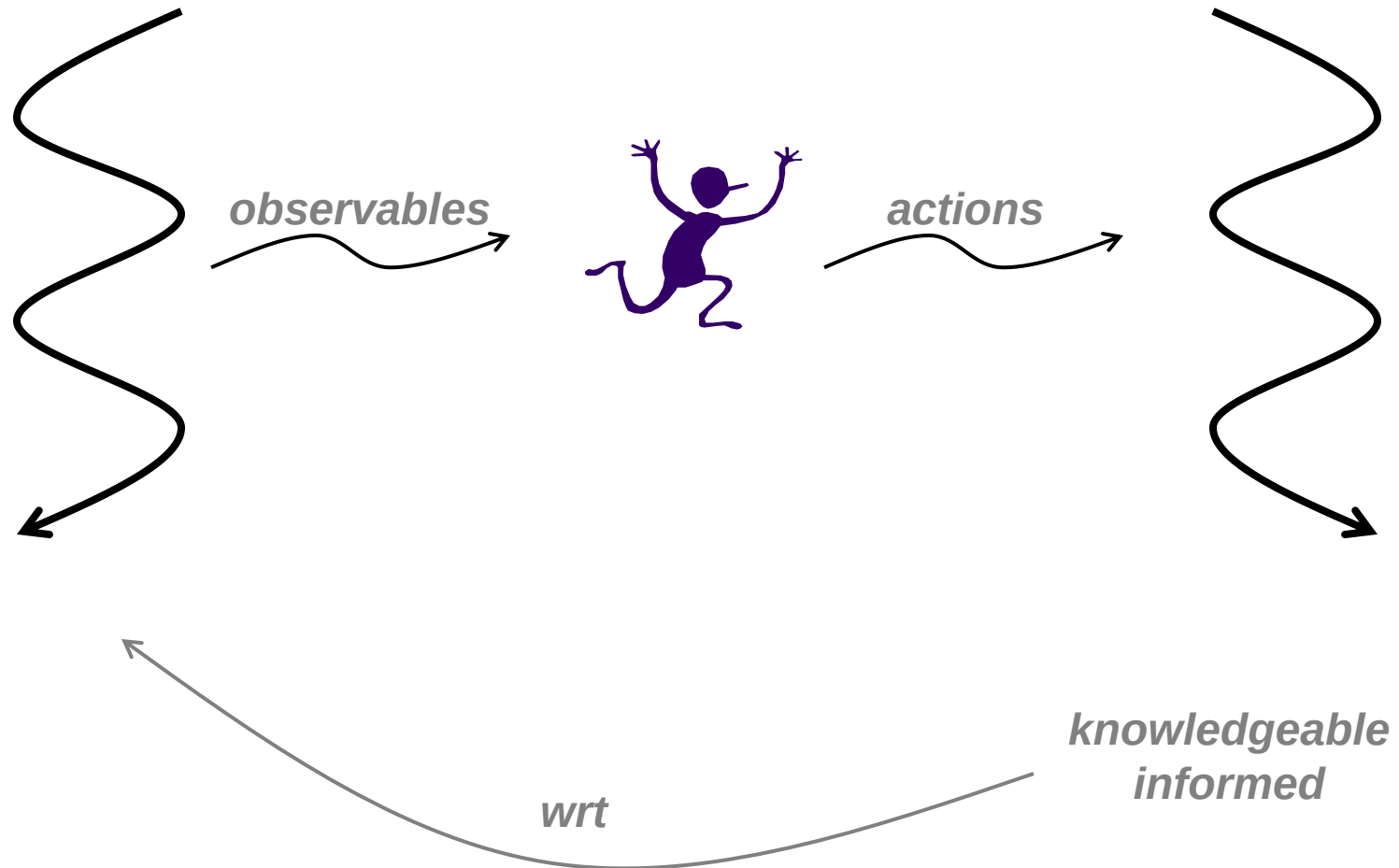
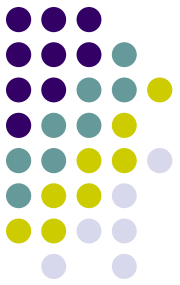


Which concepts are (at least) needed to model behaviour?

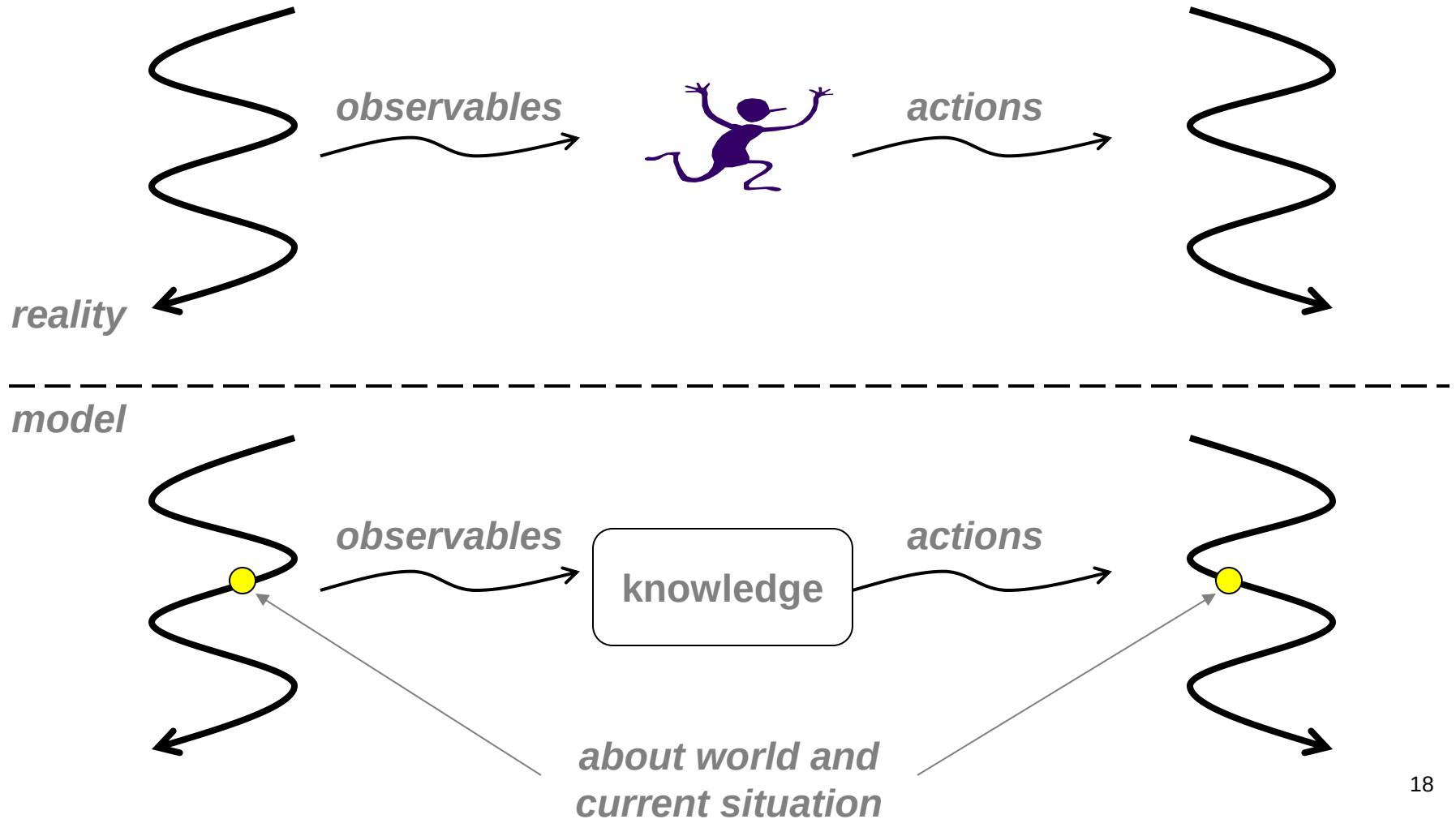
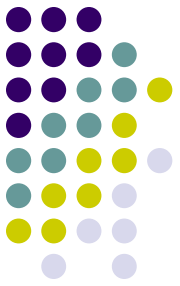
Behaviour



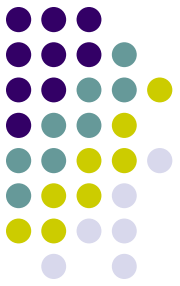
Intelligent behaviour



Modelling intelligent behaviour

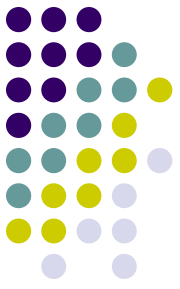


Side path: what do we model?



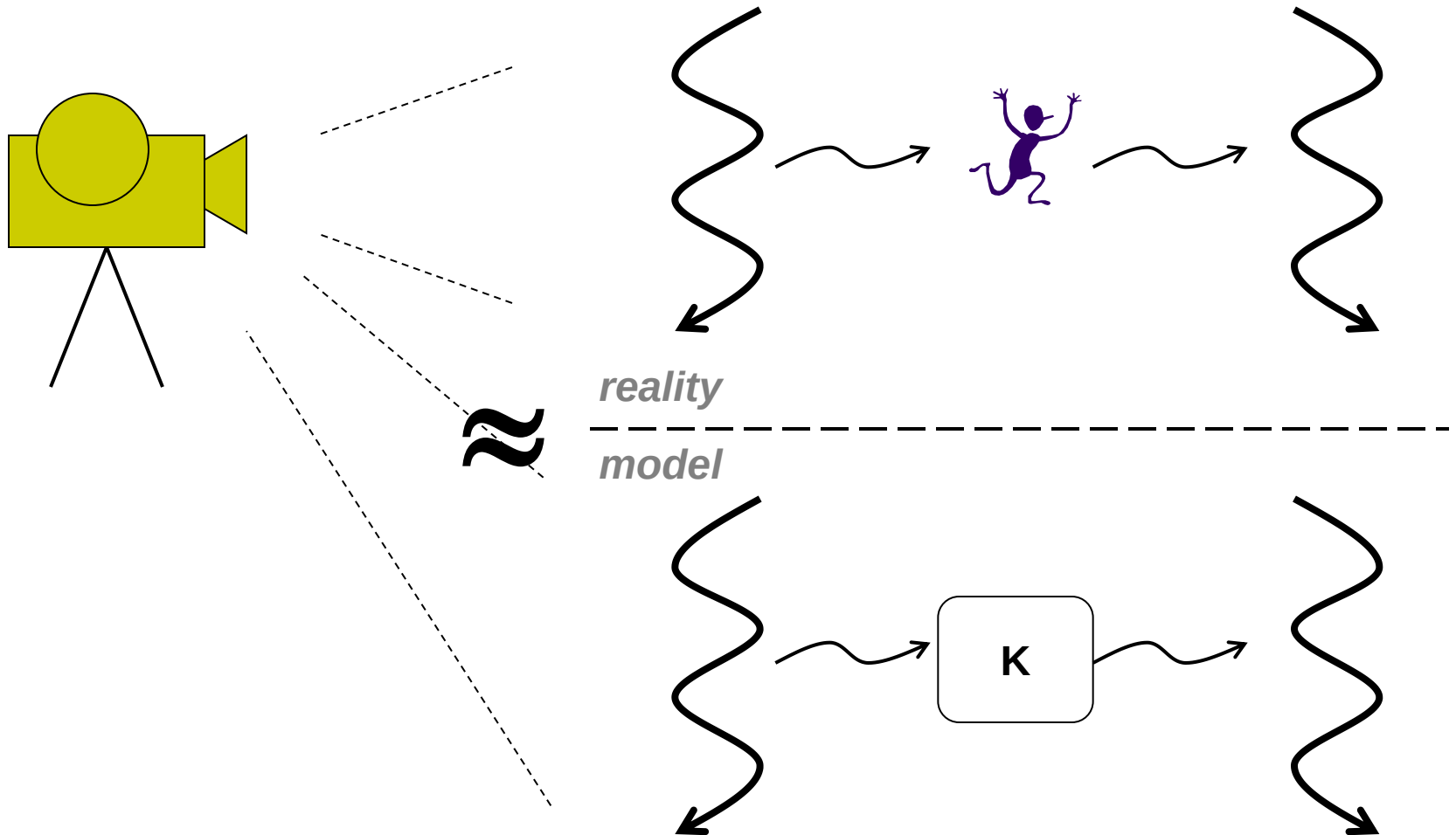
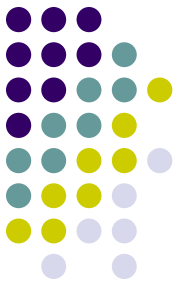
- What is the **meaning** of this “knowledge that describes behaviour”?
 - is it just a description of what is observed?
 - is it an explanation?
 - is it a cognitive model, describing the working of our brain?

Two perspectives on modelling behaviour

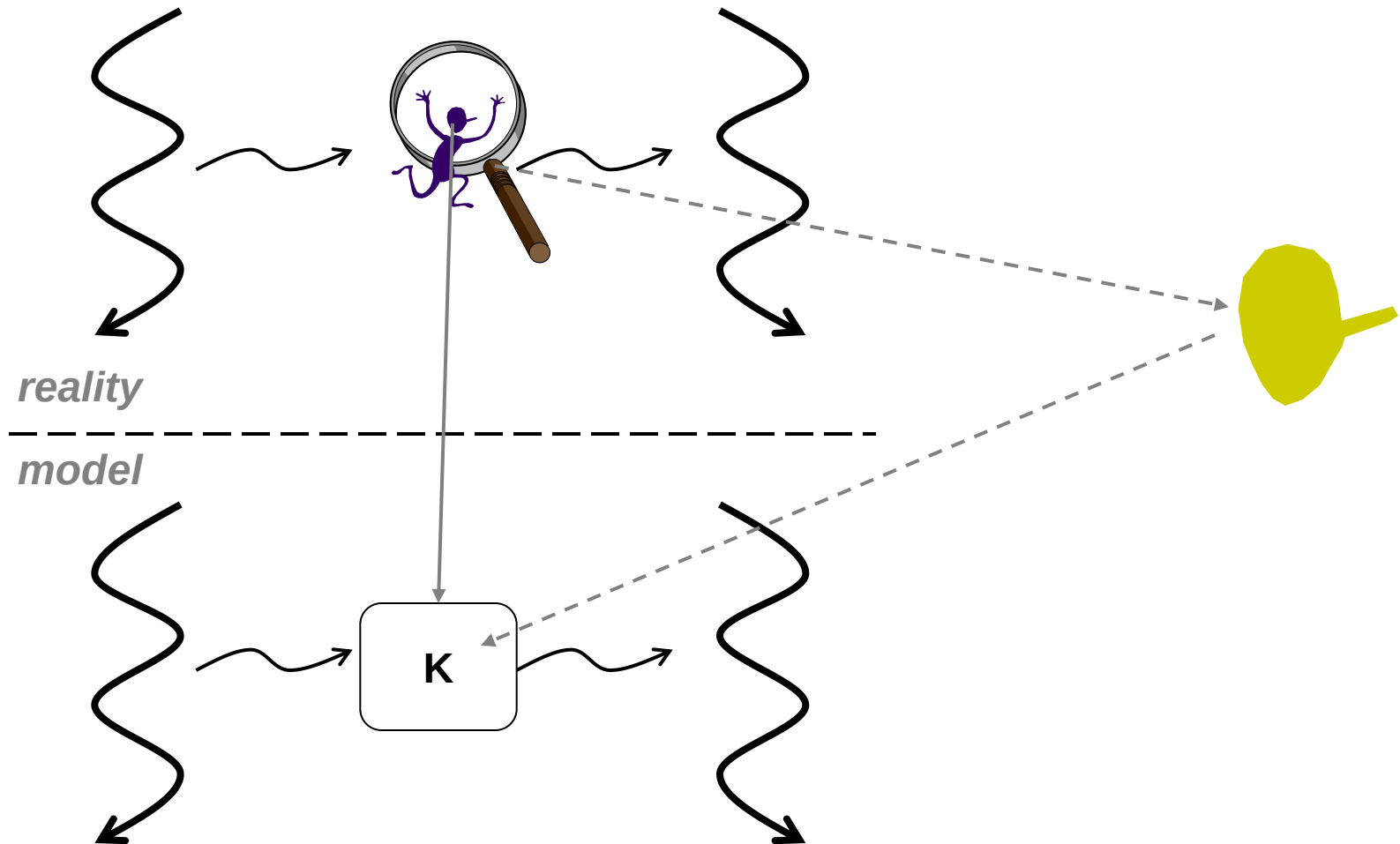
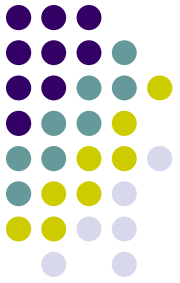


- External (observers) perspective
 - knowledge exploited as a useful means to **externally explain** the subject's behaviour without making any claim on the actual internal functioning of the subject.
- Internal (the subject's own) perspective:
 - a subject actually has knowledge, encoded in one way or the other in it (cognition), and this **knowledge is used by the subject** to generate its behaviour.

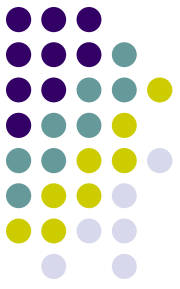
External viewpoint



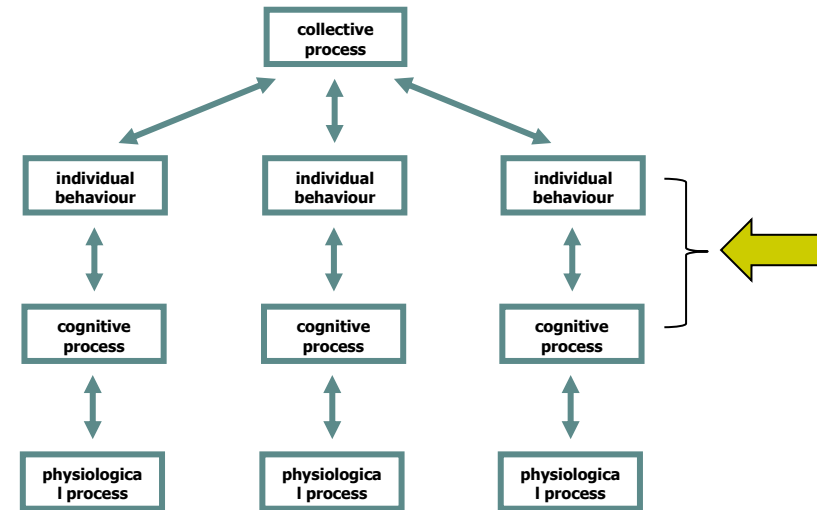
Internal Viewpoint



Central theme in cognitive science

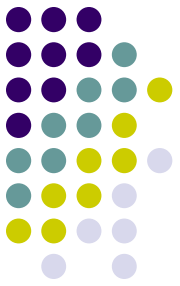


- Analyse how cognitions relate to behaviour
 - *how internal descriptions relate to external observable behaviour*



- Approach: modelling and simulation
 - externally observable behaviour is used as “characteristic patterns” to provide evidence for the feasibility of internal descriptions

Environmental Complexity Thesis (1)

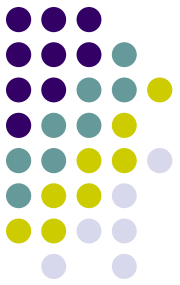


- The function of cognition is to enable the agent to deal with environmental complexity

(Godfrey-Smith, 1996)

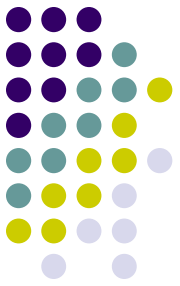
Environmental Complexity

Thesis (2)

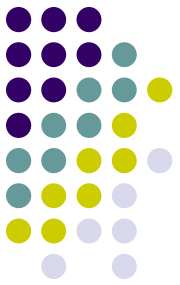


- The more **complex the environment**, the **more sophisticated the behaviour** that is required to deal with this environment
- The more sophisticated the behaviour that is required to cope with this environment the **more complex the mental representations** and capabilities needed.

What we will do for modelling behaviour

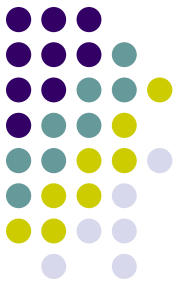


- Look at behavioural processes from an external viewpoint
 - observations, actions
- Come up model consisting of with **internal cognitive states** that generates the behavioural pattern
- Develop and learn standard **vocabulary** for cognitive models



Different behaviour patterns

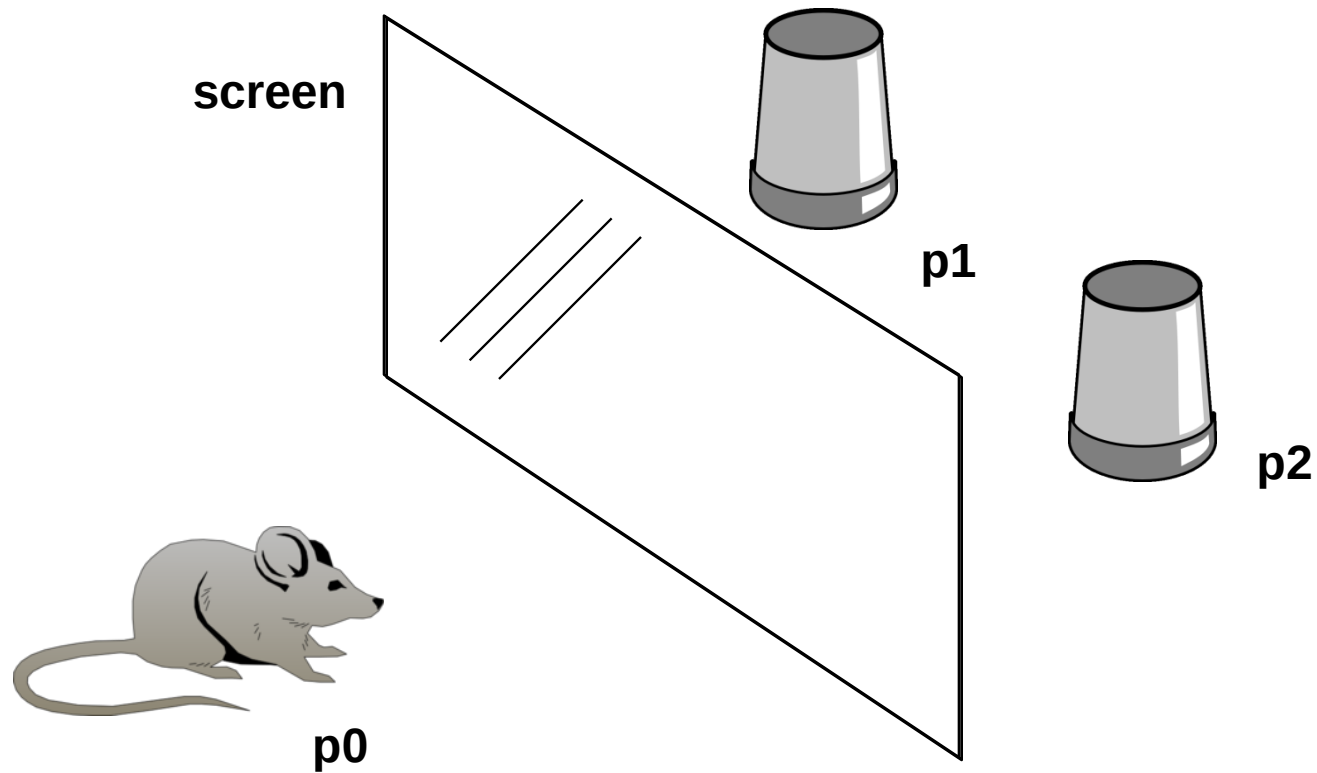
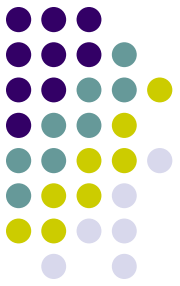
- Reactive behaviour (Chapter 9)
 - stimulus-response
 - delayed response
- Motivation-based behaviour (Chapter 10)
- Adaptive behaviour (Chapter 11)
- Mind-in-the-world behaviour (Chapter 12)

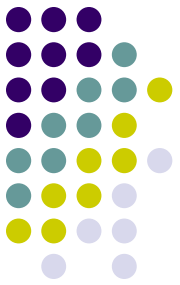


Chapter 9

REACTIVE BEHAVIOUR

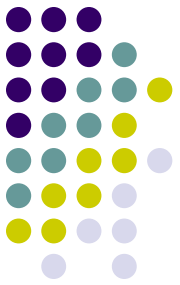
Experimental Setting





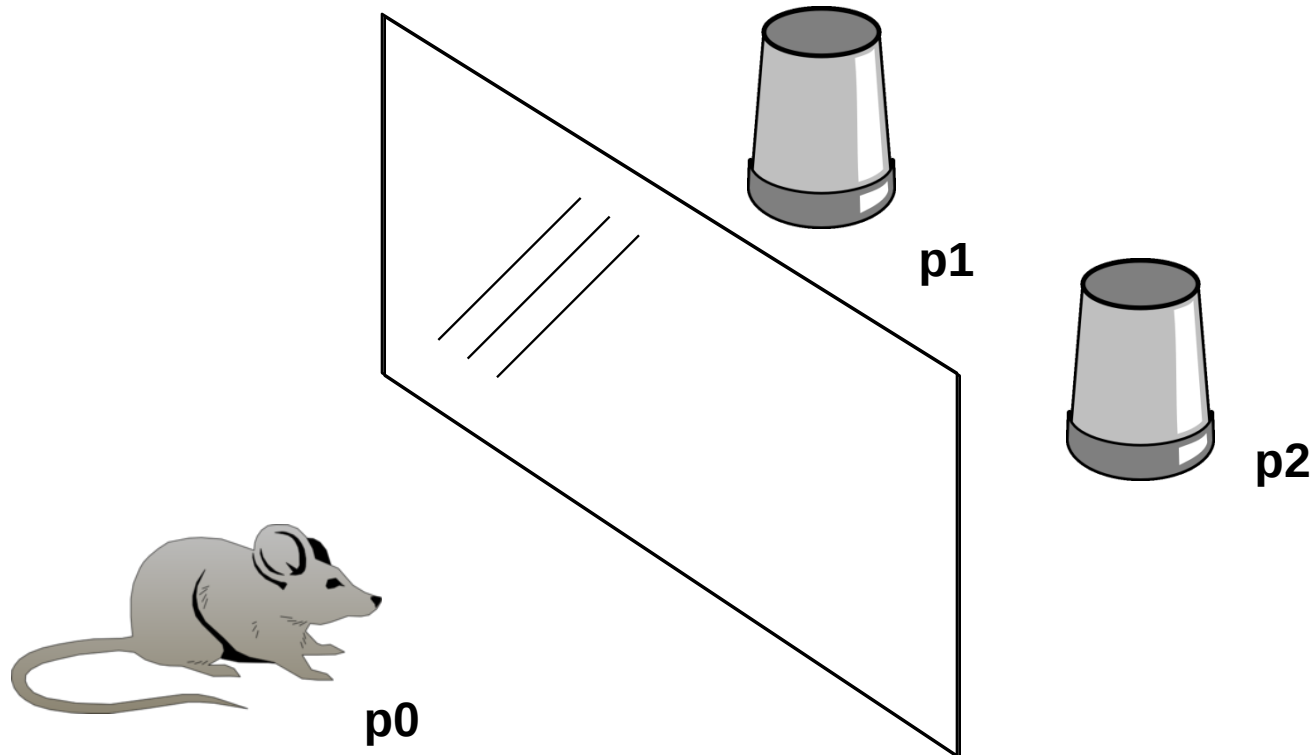
Experimental scenarios

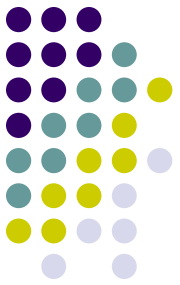
1. empty cup at **p1** and **p2**
2. empty cup at **p1**,
food (visible) at **p2**
3. empty cup at **p1**,
food at **p2**,
cup at **p2** (food no longer visible)
4. as 3. but food was never visible



Results: scenario 1

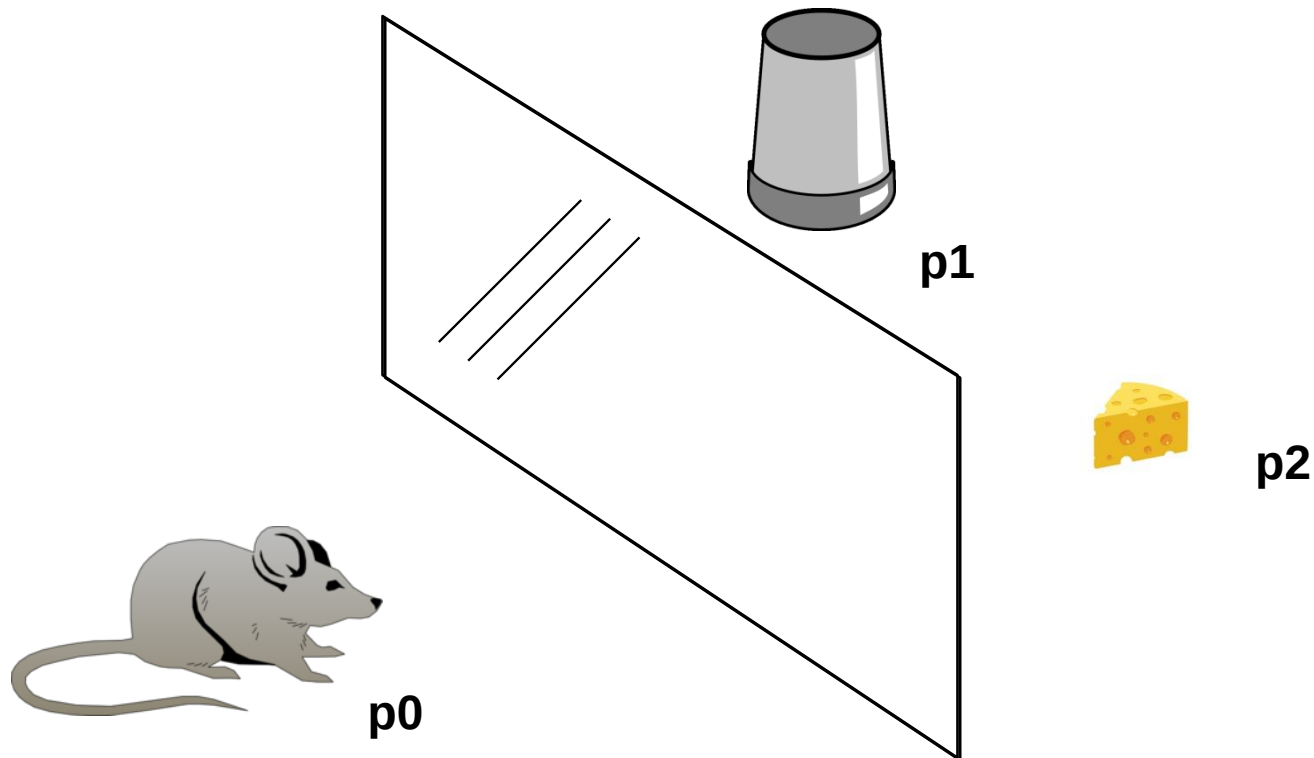
- Animal shows no preference for **p1** or **p2**.
- Animal may wander or stay at **p0**.

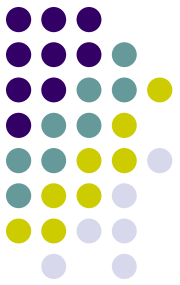




Results: scenario 2

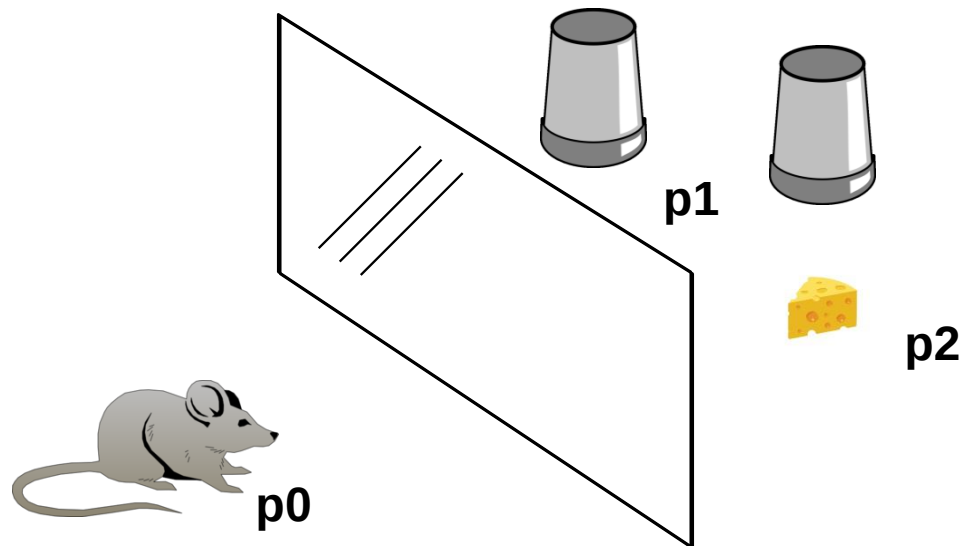
- Animal goes to **p2**
(explanation: stimulus-response behaviour)

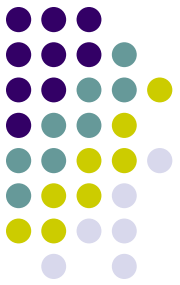




Results: scenario 3

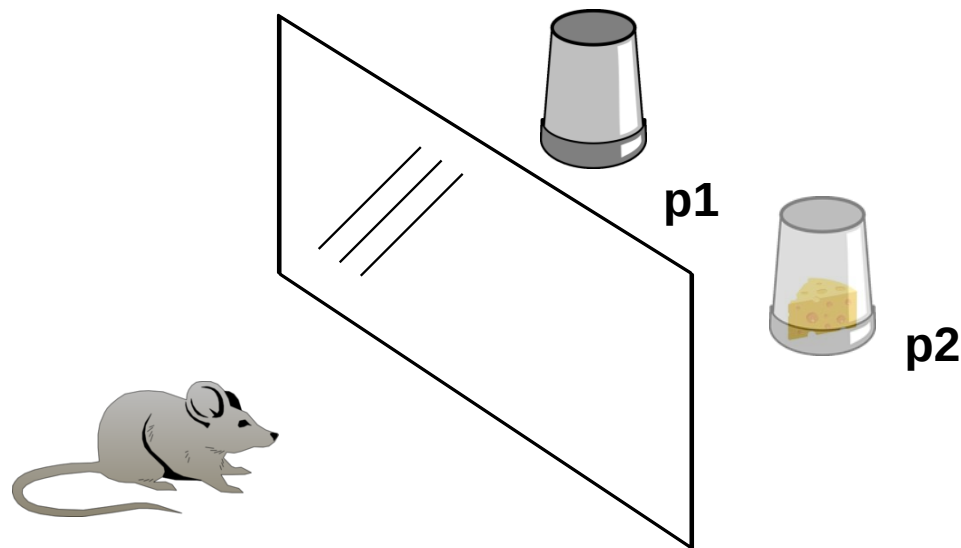
- Different possibilities:
 - Some animals behave as in situation 1
 - Some animals go to p2
(explanation: **delayed response behaviour**)

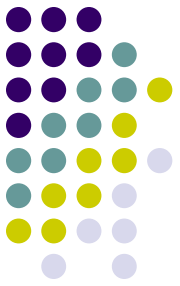




Results: scenario 4

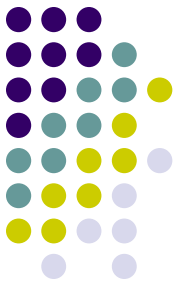
- Most animals behave as in situation 1, but:
 - some animals may search food (motivation-based behaviour)
 - some animals may find food, and learn (adaptive behaviour)





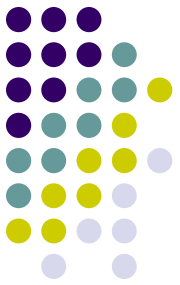
Types of behaviour

- Type A: stimulus-response, no delayed response
- Type B: delayed response
- Type C: motivation-based
 - behaviour depends on motivation or goal
 - may act pro-actively in situation 1 (hungry)
 - may ignore food stimuli in situation 2
- Type D: adaptive



Assignment

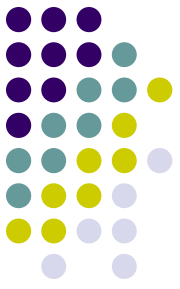
- Think about human behaviour.
Write down one example of either :
 - stimulus-response behaviour
 - delayed-response behaviour
 - motivation-based behaviour
 - adaptive behaviour



Today: reactive behaviour

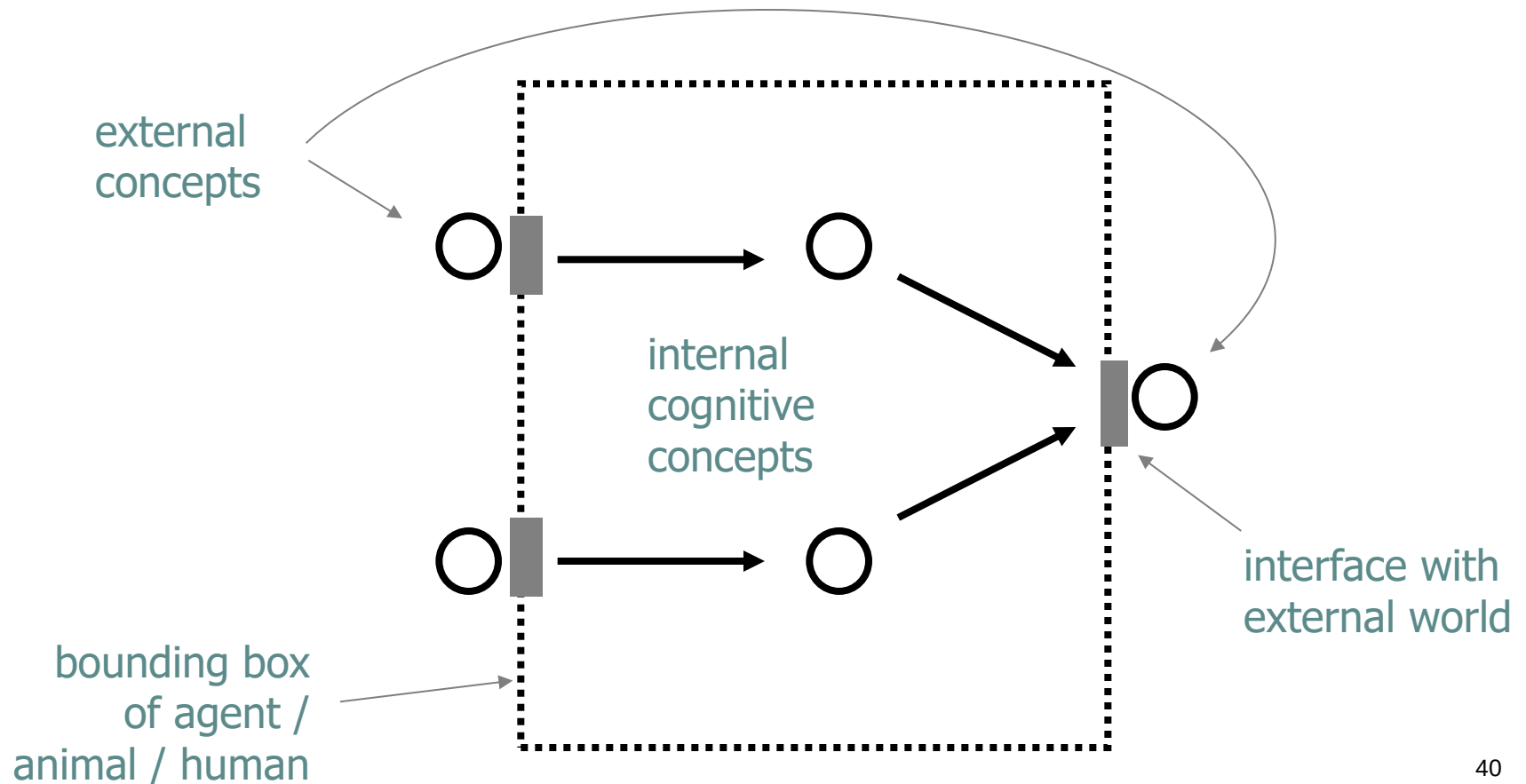
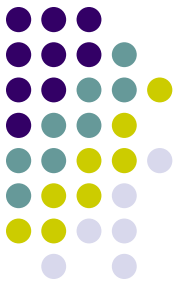
- Includes the following two types:
 - stimulus-response behaviour
 - delayed response behaviour
- What we will do:
 - try to describe both types of behaviour
 - from external and internal perspective
 - then formalize the models in LEADSTO (logical formalism)

Describing behavior in logical way

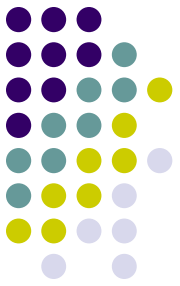


- By means of **executable dynamic properties**
 - logical relations that specify truth value dependencies between **state properties** at next time step
- During conceptualisation we use a **semi-formal notation**
 - at any point in time:
IF... THEN ... (at the next time point)

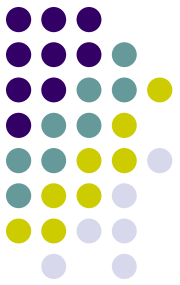
Conventions for graphical model of behaviour



Stimulus-Response Behaviour: Example Traces



<i>time trace</i>	<i>0</i>	<i>1</i>	<i>2</i>	<i>3</i>	<i>4</i>	<i>5</i>
<i>trace 1</i>	food at p2 screen	no food screen	food at p2 screen	food at p2 no screen	food at p2 no screen goes to p2	no food no screen goes to p2
<i>trace 2</i>	no food screen	no food screen	no food screen	food at p2 screen	food at p2 no screen	food at p2 no screen goes to p2
<i>trace 3</i>	no food screen	no food no screen	no food no screen	food at p2 no screen	food at p2 no screen goes to p2	no food no screen goes to p2
<i>trace 4</i>	food at p2 screen	no food no screen	food at p2 screen	food at p2 screen	food at p2 no screen	food at p2 no screen goes to p2



Description of behaviour

- From an external perspective:

at any point in time

IF the animal observes food at p2

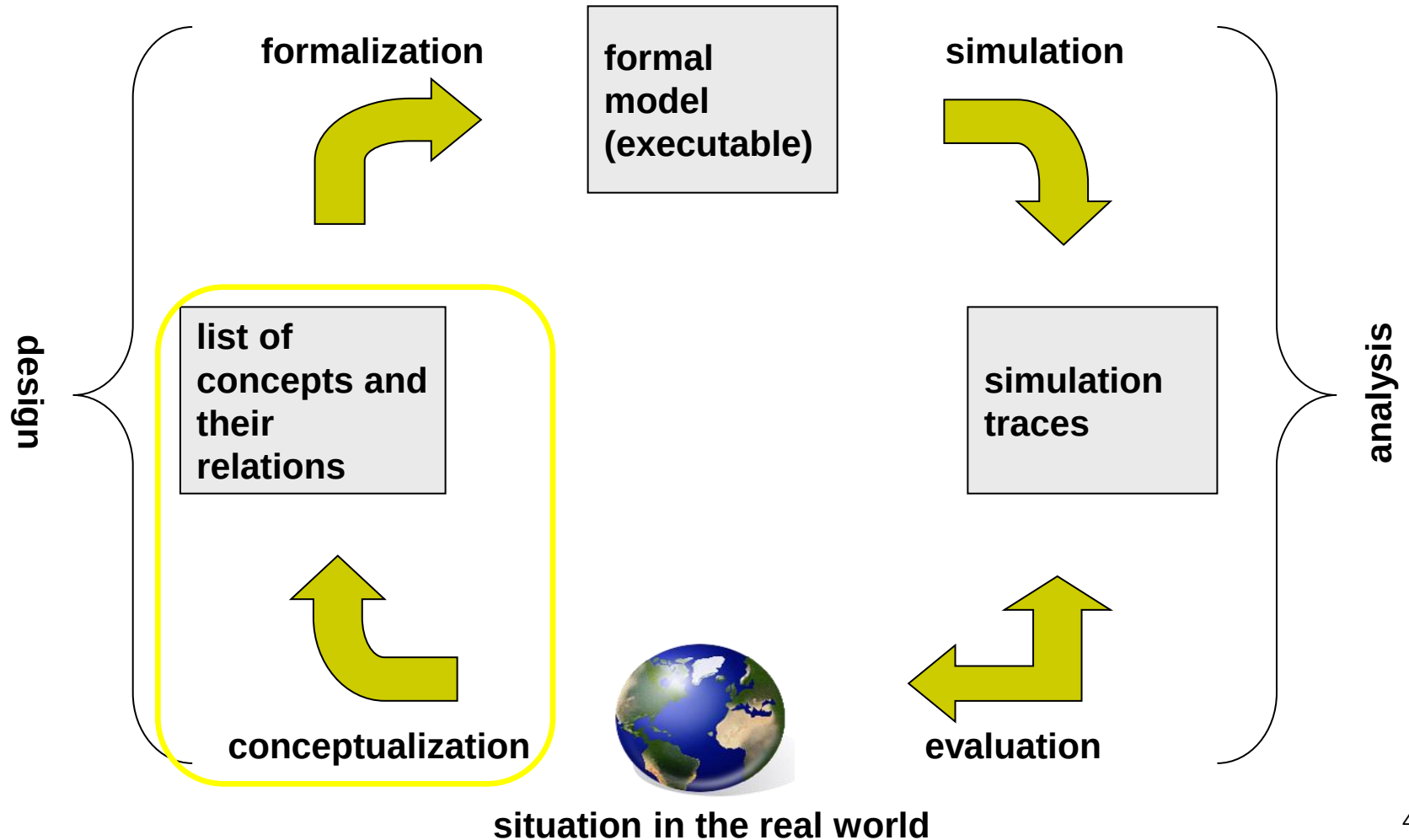
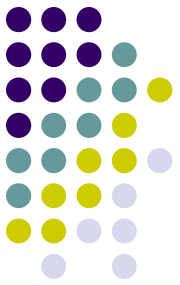
AND the animal observes

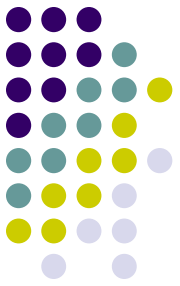
that there is no screen

THEN the animal will go to p2

(at the next time point)

Modelling the Behaviour

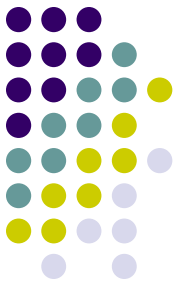




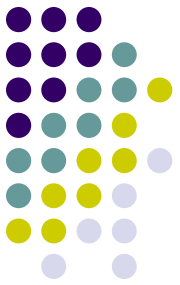
Conceptualization

- Make a list of what are considered:
 - input concepts and
 - output concepts
- for an organism performing SR behaviour

Specification of input and output



- Input: observation
(of presence of food and screen)
- Output: action
(of going to p2)



Relations between concepts

- Stimulus-response behaviour:

An observation that there is food at p2

affects

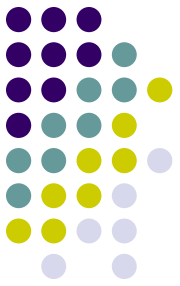
that it will go to p2.

An observation that no screen is present

affects

that it will go to p2.

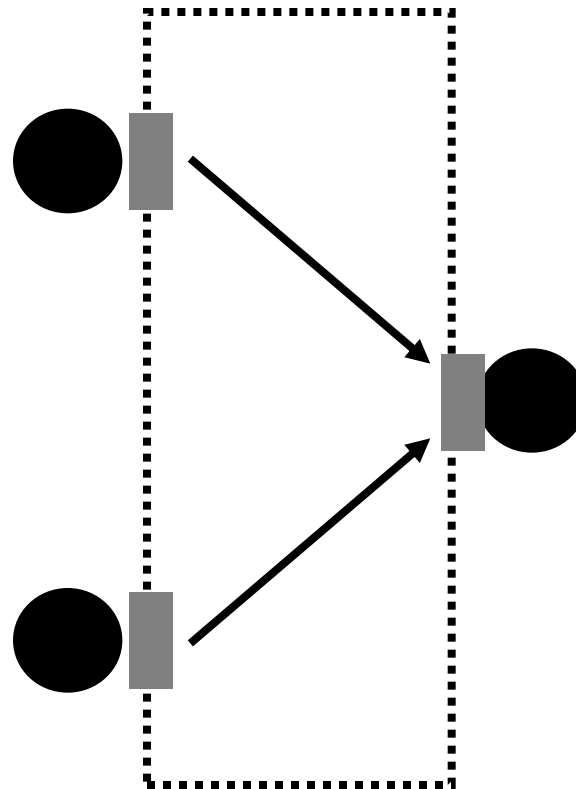
Relationships between Concepts



- Stimulus-response behaviour:

**observes
food at p2**

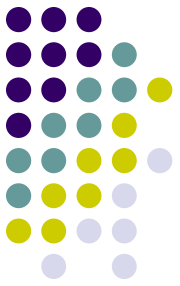
**observes
no screen**



goes to p2

QUESTION:

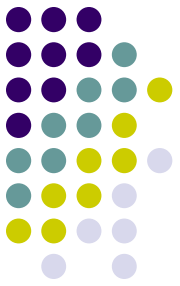
*Is this a description
from an internal or an
external viewpoint?*



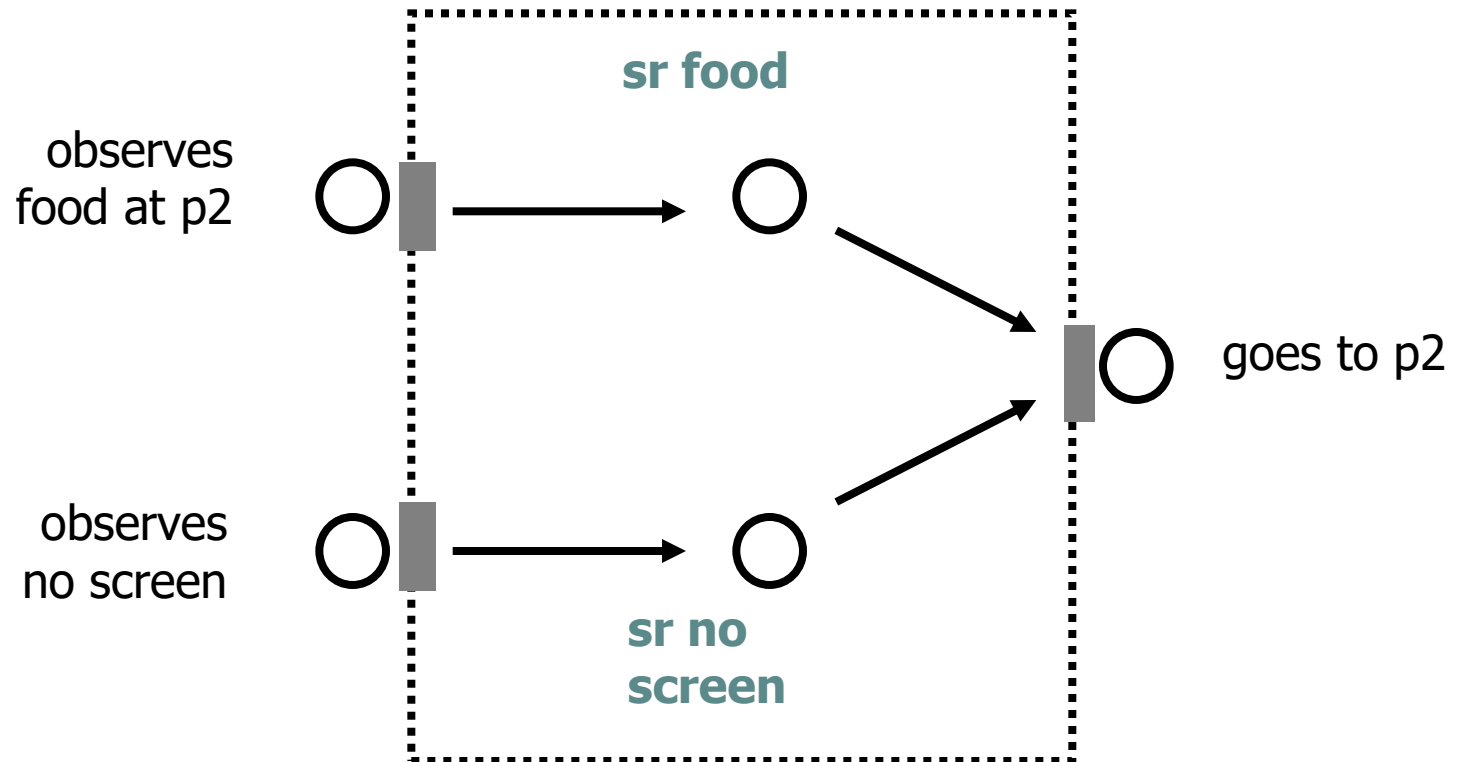
Internal viewpoint

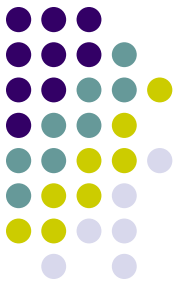
- What would be needed from an internal viewpoint?
 - specification of internal cognitive states
- Internal: sensory representation
(of presence of food and screen)

Relationships between concepts



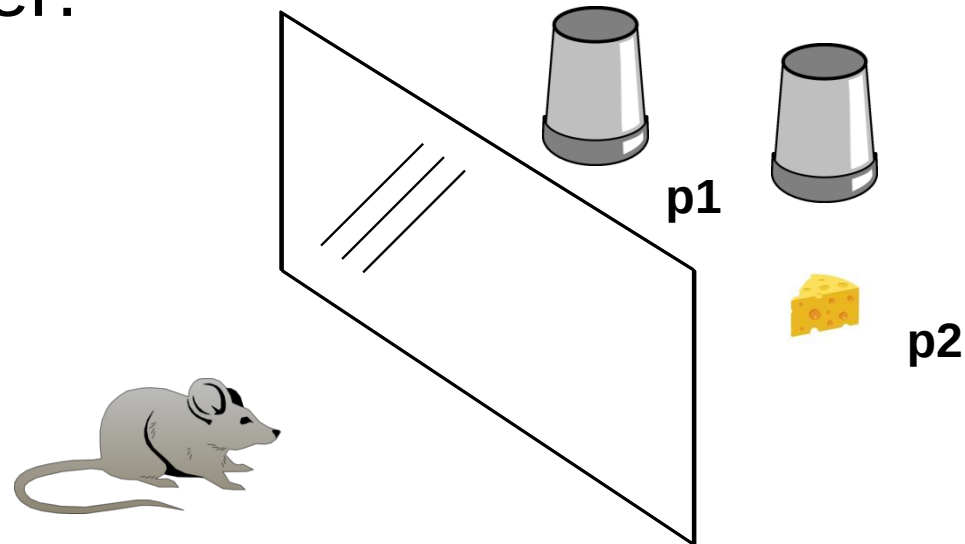
- Stimulus-response behaviour:



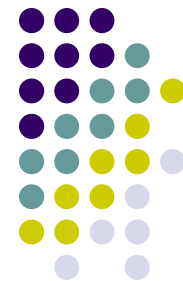


Delayed Response Behaviour

- A more sophisticated form of reactive behaviour.
- Remember:

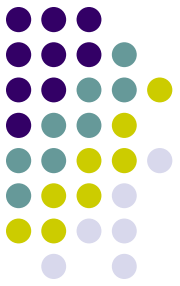


Delayed-response behaviour: example traces



	<i>0</i>	<i>1</i>	<i>2</i>	<i>3</i>	<i>4</i>	<i>5</i>
<i>trace 1</i>	food screen	cup screen	cup screen	cup no screen	cup no screen goes to p2	cup no screen goes to p2
<i>trace 2</i>	no food screen	no food screen	cup screen	cup screen	cup no screen	cup no screen
<i>trace 3</i>	no food screen	no food no screen	no food no screen	no food no screen	food no screen	food no screen goes to p2
<i>trace 4</i>	food screen	no food screen	food screen	cup screen	cup no screen	cup no screen goes to p2

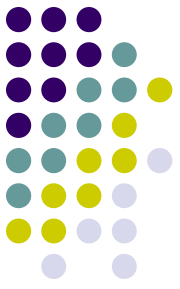
Delayed-response behaviour: external description



FOR all TIMEPOINTS,
if the animal observes that no screen is
present
and at some earlier point in time
the animal observed that food was present
at position p2,
then the animal will go to position p2

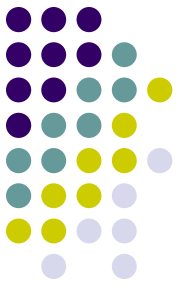
Rigid form: no update when observed that food is removed!

Delayed-response behaviour: alternative description



at any point in time t_1 ,
if the animal observes that no screen is
present
and at some earlier point in time t it observed that
food was present at position p_2 ,
and at every point in time t' after t up to t_1 ,
it did not observe that no food was present
at p_2 ,
then the animal will go to position p_2

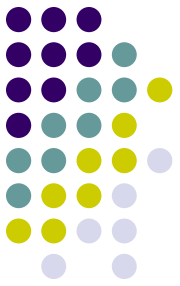
Updating form



Internal cognitive states

- How do we represent this at an internal cognitive level?
- Internal: **belief**
(of presence of food and screen)

Relationships between concepts



- Delayed-response behaviour:

an observation about food at p2

affects

a belief about food at p2.

an observation that no screen is present

affects

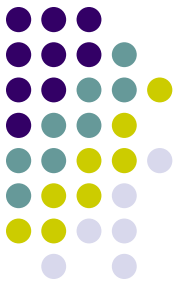
that it will go to p2.

a belief about food at p2.

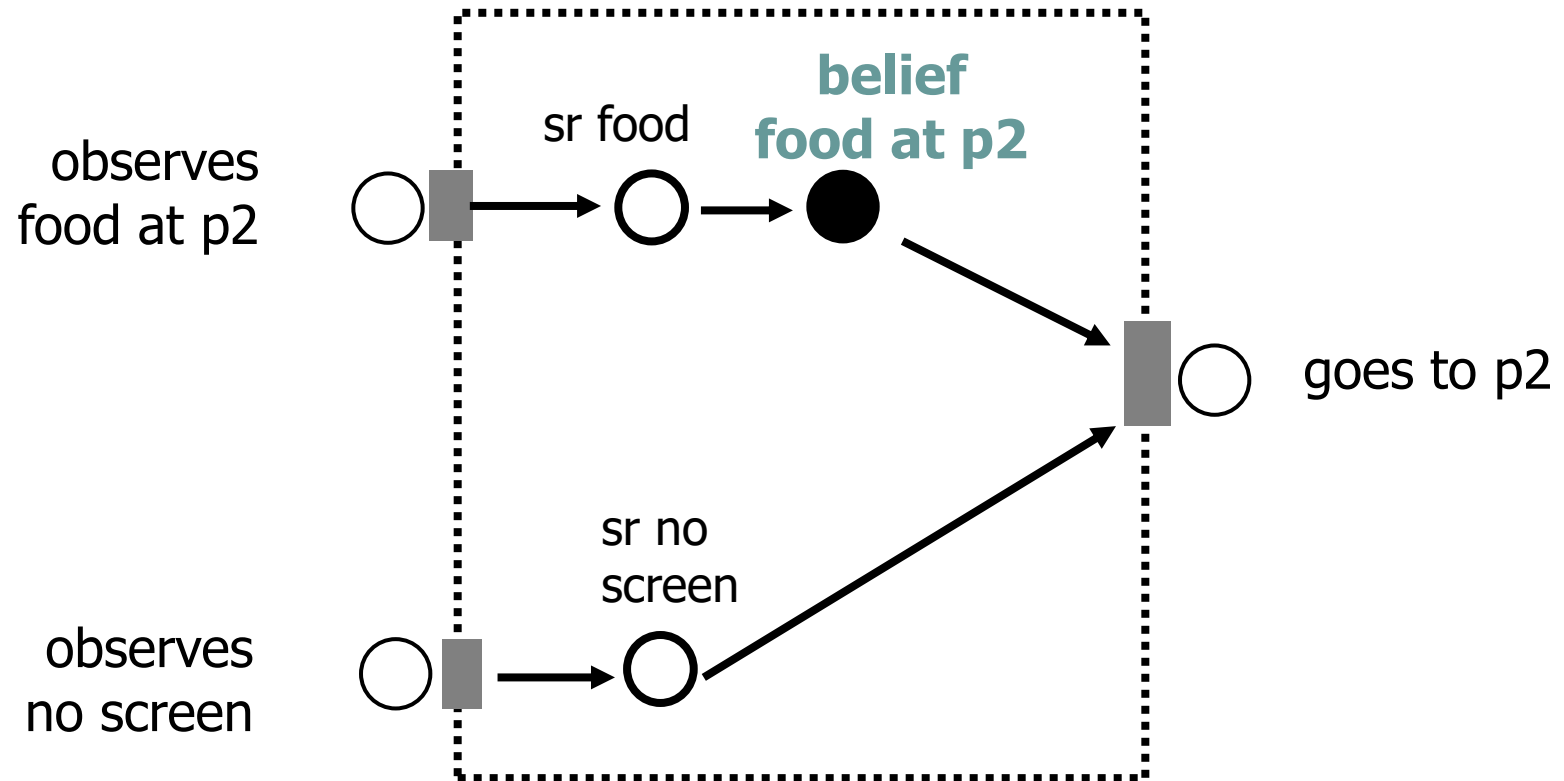
affects

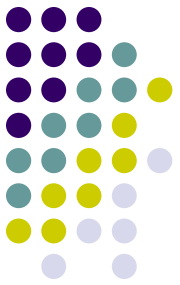
that it will go to p2.

Relationships between Concepts

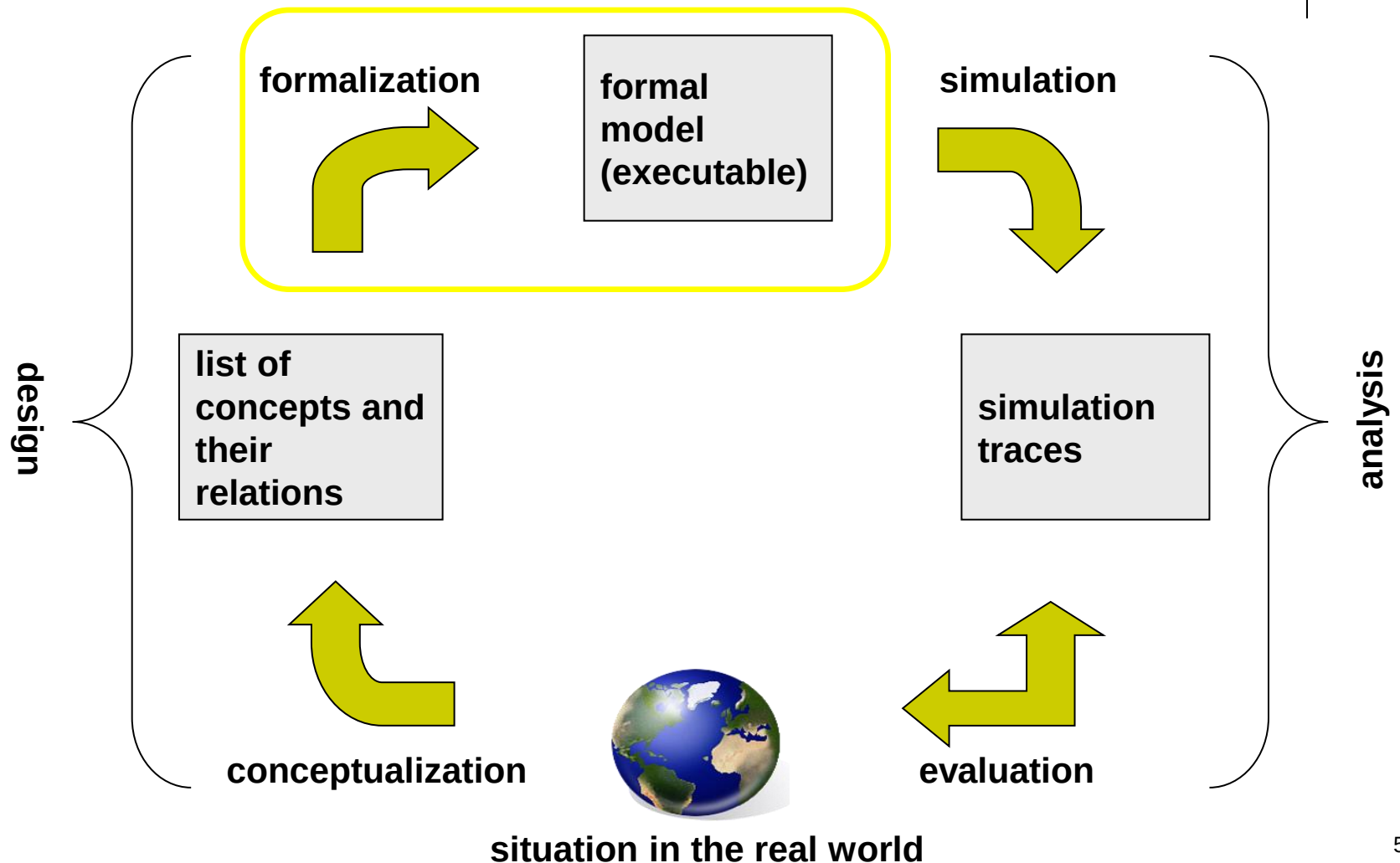


- Delayed Response Behaviour

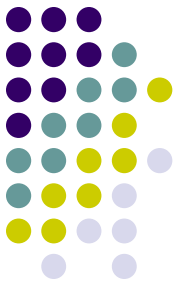




Modelling Cycle: Formalization



Formalisation of concepts in Leads2



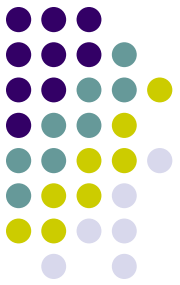
- Logical statements that can have arguments:

`observed(_)`

`sensory_representation(_)`

`at_position(_,_)`

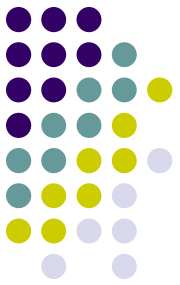
- Value of statement can be:
 - **true**, **false** or **unknown**



The detailed model

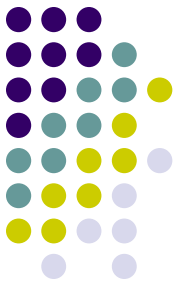
- Formalisation of the concepts:

External behavioural level	State properties
observation (of presence of food) (of presence of screen)	observed(at_position(food, p2)) observed(screen)
action (goto p2)	performed(goto(p2))
Internal cognitive level	
sensory representation (of presence of food) (of presence of screen)	sr(at_position(food, p2)) sr(screen)
belief (of presence of food) (of presence of screen)	belief(at_position(food, p2)) belief(screen)



Formalizing logical relations

- Remember: **executable dynamic properties**
 - logical relations that specify truth value dependencies between **state properties** at next time step
- Notation:
 - semi-formal notation (if ... then at next time step ...)
 - formal notation ($\rightarrow\!\!\rightarrow$)



Example LEADSTO relation

Executable dynamic property

If X holds
then Y will hold

“LEADSTO” relation

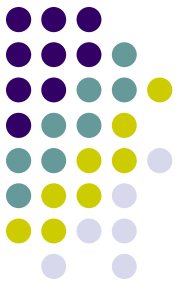


X, Y (combinations of) state properties

$X \longrightarrow Y$

$X1 \wedge X2 \longrightarrow Y$

Warning: facts always need to be derived explicitly



- No closed-world assumption
 - if no truth value is generated for a predicate, there is no information about state in next time step
- Therefore, two types of dynamic properties:

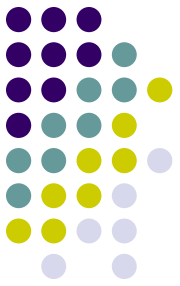
Step Property (new facts in next state)

If X holds $X \longrightarrow Y$
then Y will hold

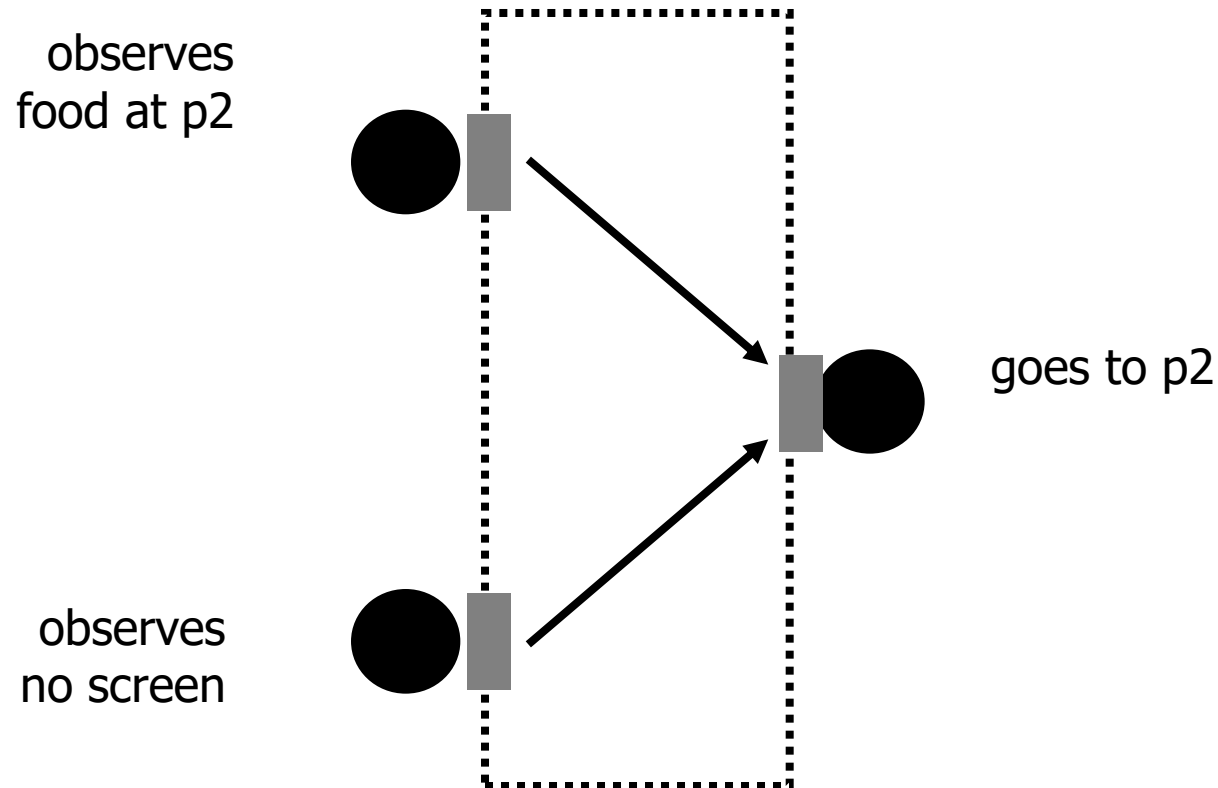
Persistence property (maintaining facts in next state)

If X holds
(and Y does not hold) $X \wedge \neg Y \longrightarrow X$
then X will hold

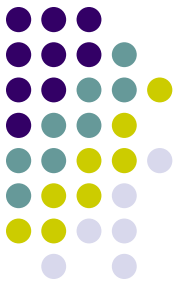
Relationships between concepts



- Stimulus-response behaviour:



Stimulus-response behaviour: External dynamics



External Dynamics:

ESR1

at any point in time,

if the animal observes that **food** is present at p2

and it observes that **no screen** is present,

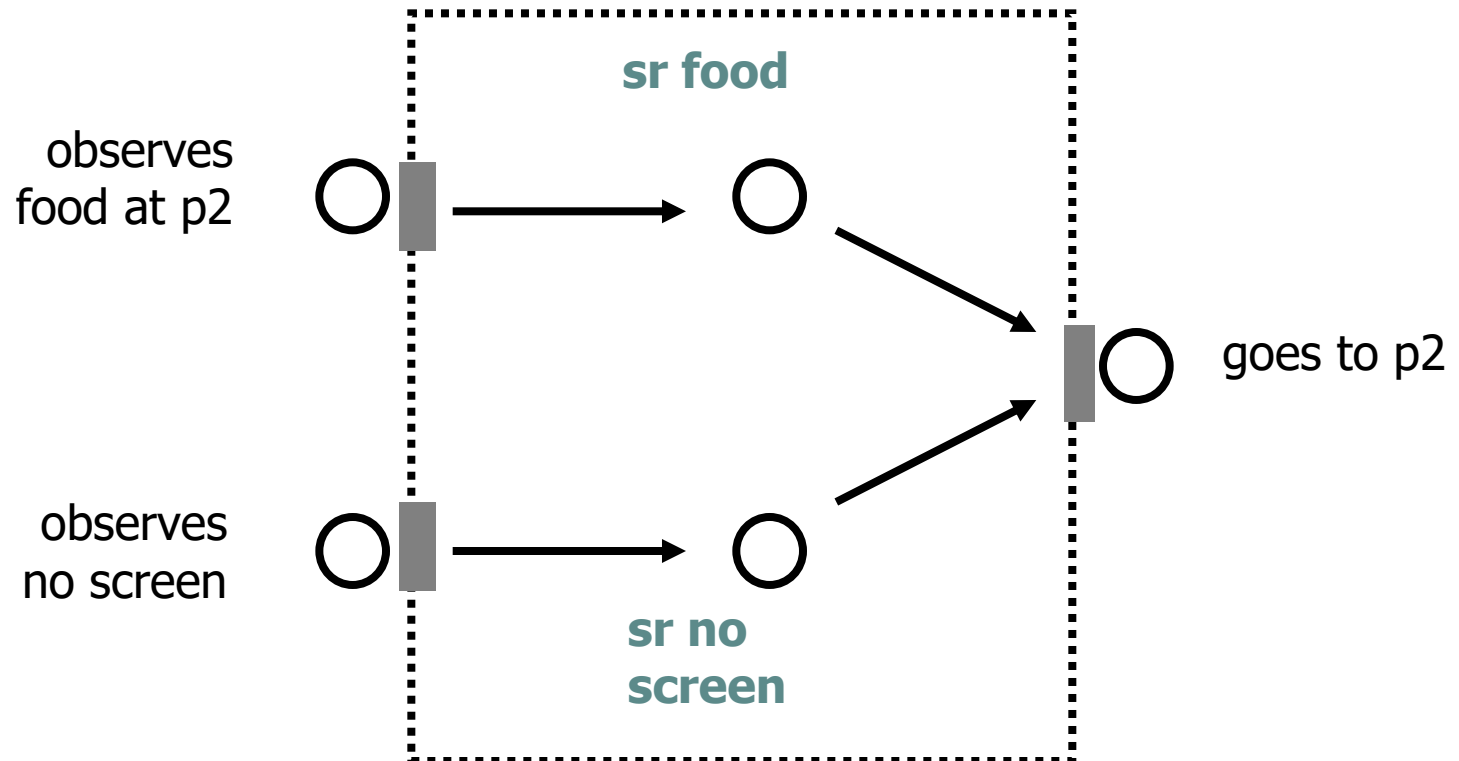
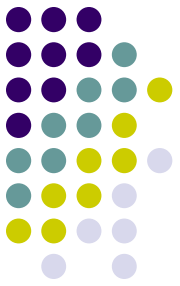
then it will **go to position p2**

$\text{observed}(\text{at_position}(\text{food}, \text{p2})) \wedge \text{observed}(\text{no_screen})$

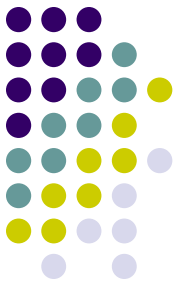


$\text{performed}(\text{goto}(\text{p2}))$

Relationships between concepts: internal perspective



Stimulus-response behaviour: Internal dynamics (1)



ISR1

at any point in time,

if the animal observes that food is present at p2

then internal sensory representation for food at p2 will hold

$\text{observed}(\text{at_position}(\text{food}, \text{p2})) \longrightarrow \text{sr}(\text{at_position}(\text{food}, \text{p2}))$

ISR2

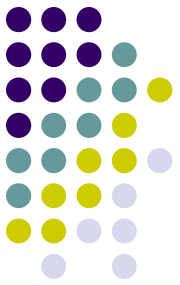
at any point in time,

if the agent observes that no screen is present,

then internal sensory representation for no screen will hold

$\text{observed}(\text{no_screen}) \longrightarrow \text{sr}(\text{no_screen})$

Stimulus-response behaviour: Internal dynamics (2)



ISR3

at any point in time,

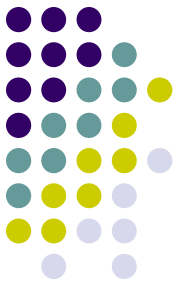
if internal sensory representation for food at p2 holds
 and internal sensory representation for no screen holds,
then the animal will go to position p2

$sr(at_position(food, p2)) \wedge sr(no_screen)$



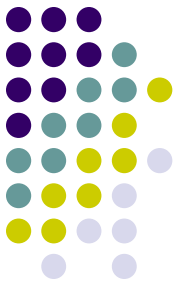
$performed(goto(p2))$

Delayed-response behaviour: external perspective

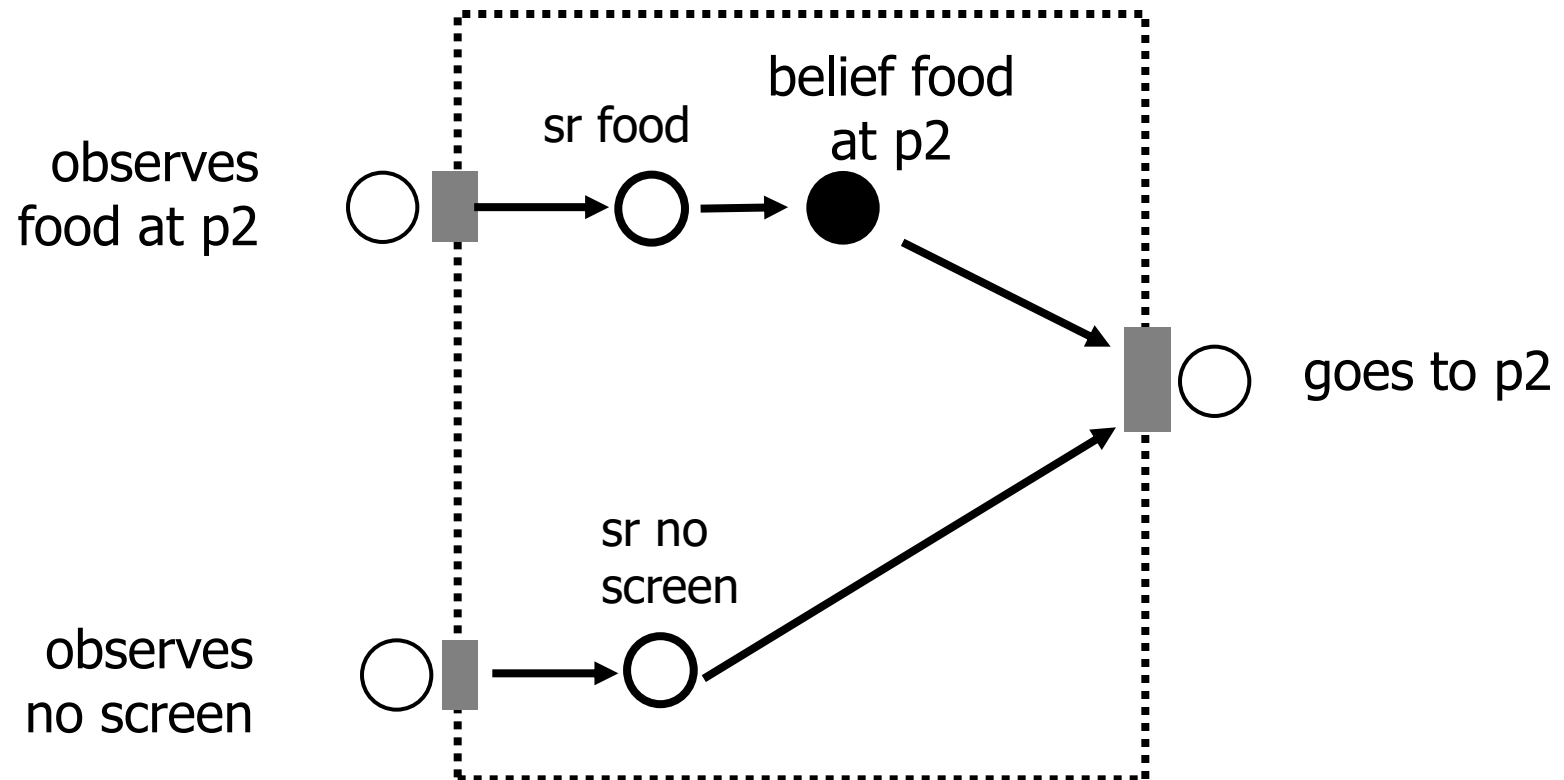


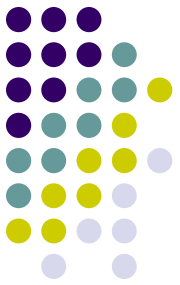
- Delayed-response behaviour
(External Dynamics):
- ??? Impossible ???

Delayed-response behaviour: internal perspective



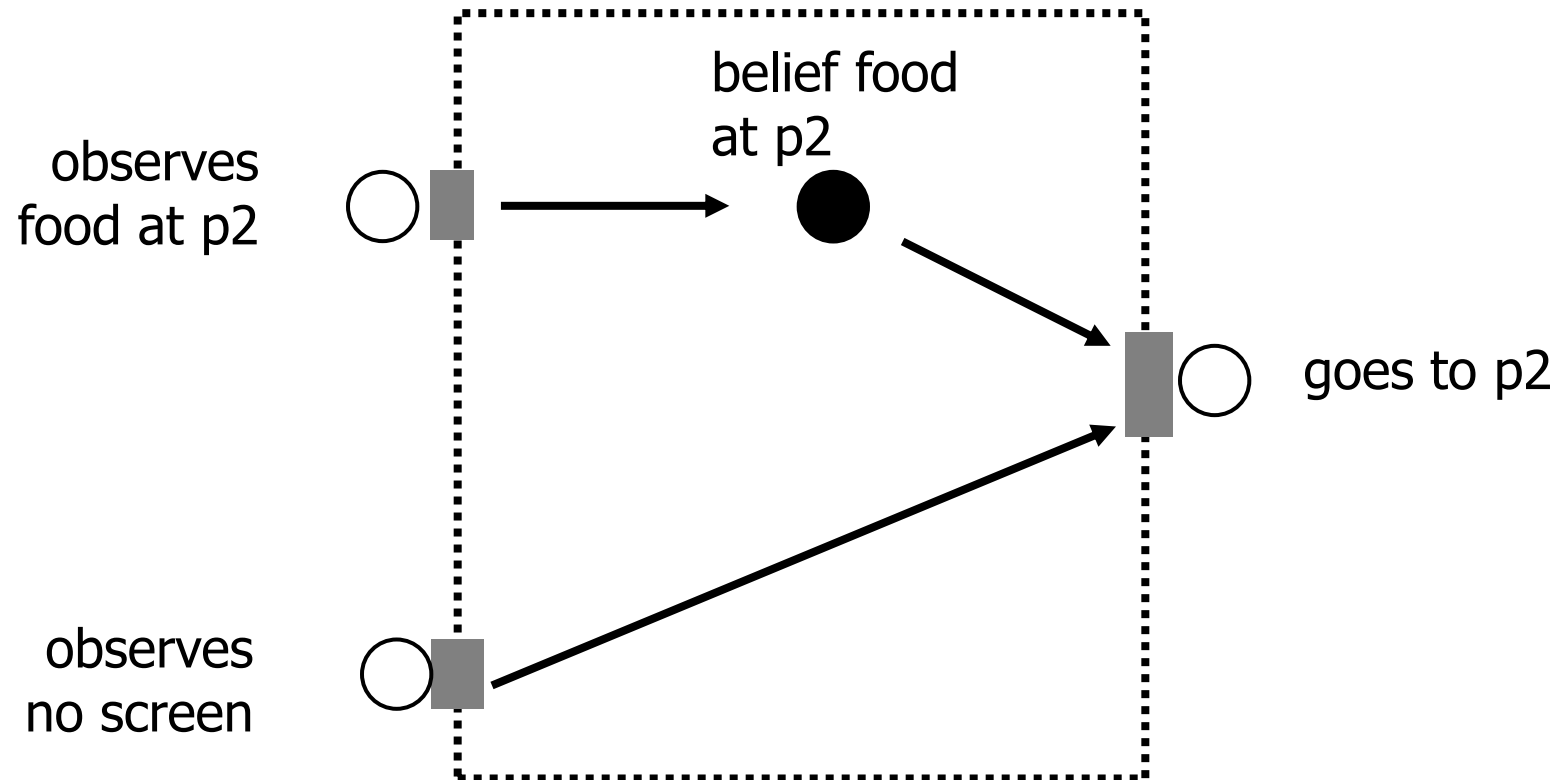
- Delayed-response behaviour
(Internal Dynamics):



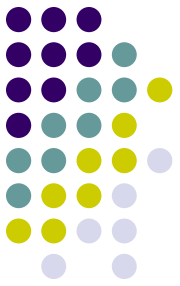


Simplified version

- Delayed-response behaviour:



Delayed-response behaviour: Internal dynamics (1)



IDR1

for all time points

if the animal observes that food is present at p2,
then internal belief that food is present at p2 will hold

$\text{observed}(\text{at_position}(\text{food}, \text{p2})) \longrightarrow \text{belief}(\text{at_position}(\text{food}, \text{p2}))$

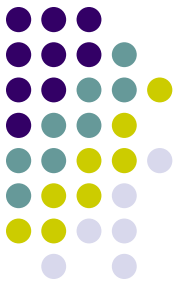
IDR2

for all time points

if internal belief that food is present at p2 holds,
then for every later time point this belief holds

$\text{belief}(\text{at_position}(\text{food}, \text{p2})) \longrightarrow \text{belief}(\text{at_position}(\text{food}, \text{p2}))$

Delayed-response behaviour: Internal dynamics (2)



IDR3

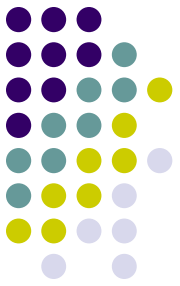
at any point in time,
if the animal observes that **no screen** is present,
and internal **belief** that food is present at p2 holds,
then the animal will **go to position p2**

observed(no_screen) $\wedge$ belief(at_position(food, p2))



performed(goto(p2))

Delayed-response behaviour: From rigid to updated



IDR2

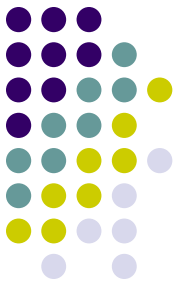
for all time points

if internal **belief** that food is present at p2
holds,

then for every later time point this **belief** holds

belief(at_position(food, p2)) $\longrightarrow$ **belief(at_position(food, p2))**

Delayed-response behaviour: From rigid to updated



IDR4

for all time points t_1 and t_2 with $t_1 < t_2$

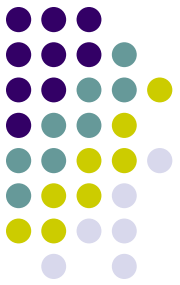
if internal **belief** that food is present at p_2 holds,
and between t_1 and t_2 the animal does not observe
that food is not present at p_2

then for every later time point this **belief** holds

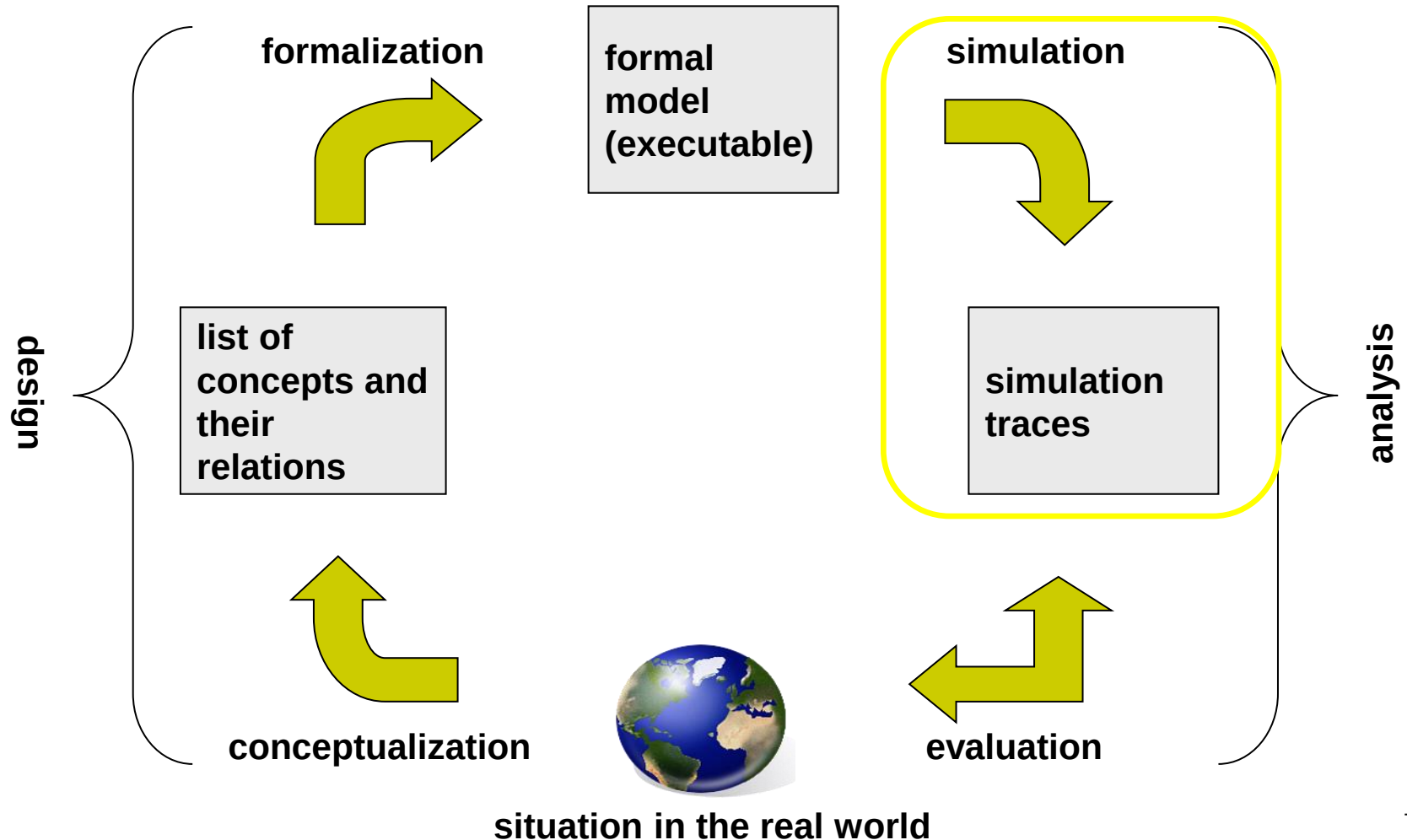
$\text{belief}(\text{at_position}(\text{food}, p_2)) \wedge \neg \text{observed}(\text{not}(\text{at_position}(\text{food}, p_2)))$

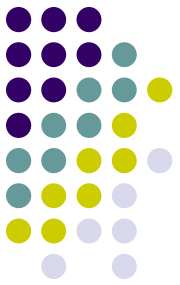


$\text{belief}(\text{at_position}(\text{food}, p_2))$



Modelling Cycle: Simulation

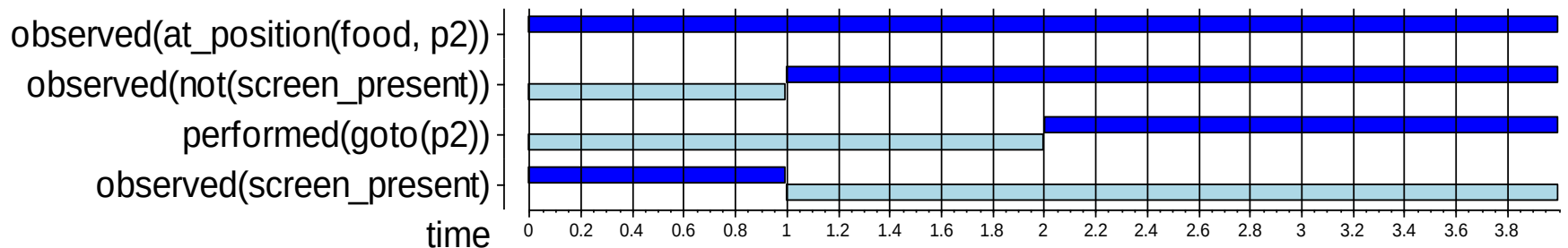
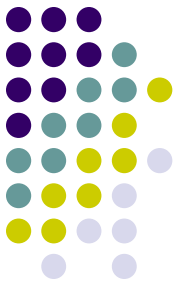




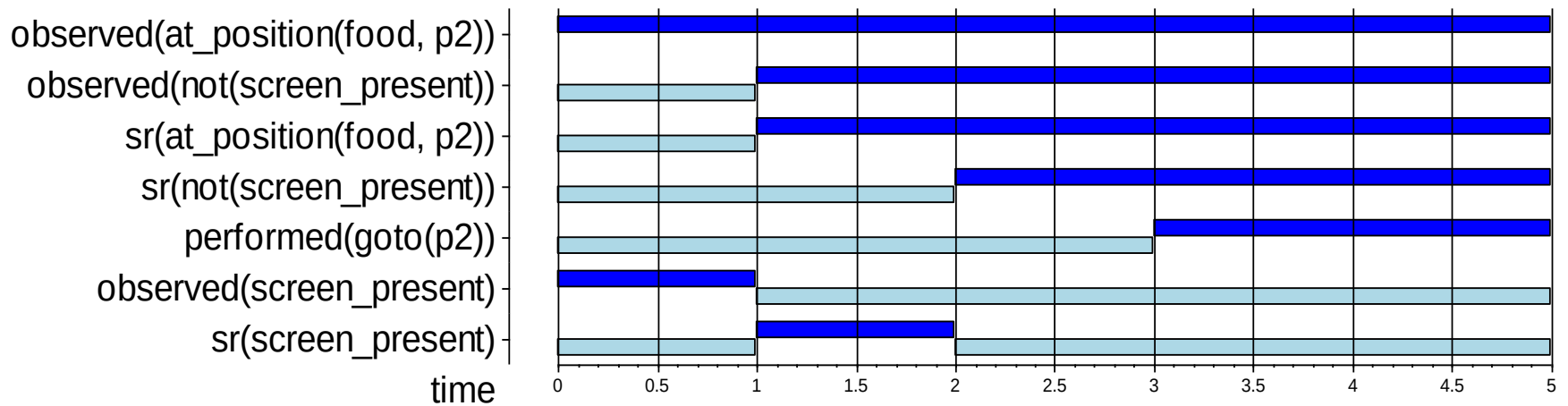
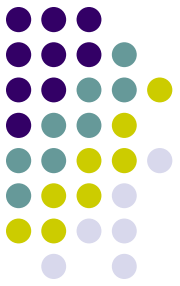
Simulation

- We use for behaviour modelling:
 - LEADSTO representation
 - Implementation in software environment
- Input: set of LEADSTO rules
- Output: simulation trace

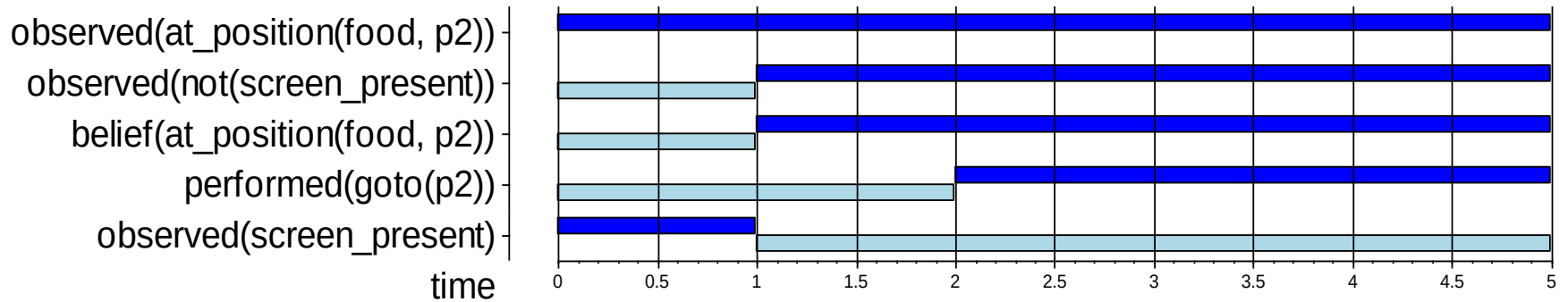
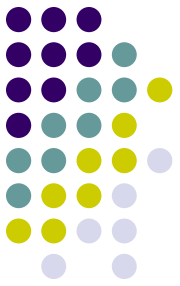
Stimulus-Response Behaviour (External)

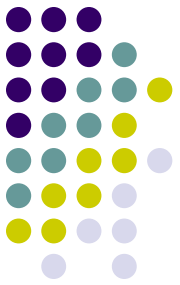


Stimulus-Response Behaviour (Internal)



Delayed Reponse Behaviour (Internal)





Take Home Message

- Modelling and simulation of behaviour requires **observations** and **actions**
- Types of behaviour:
 - stimulus-response
 - delayed response
 - motivation-based
 - adaptive
- External vs. internal viewpoint
 - external viewpoint not always possible to model
 - complex behaviour requires complex mental states!
- Logical formalism is useful representation